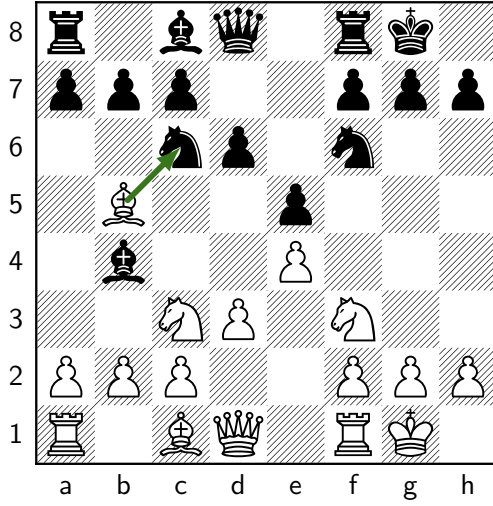


Melanie-ANALOGY — Four Knights

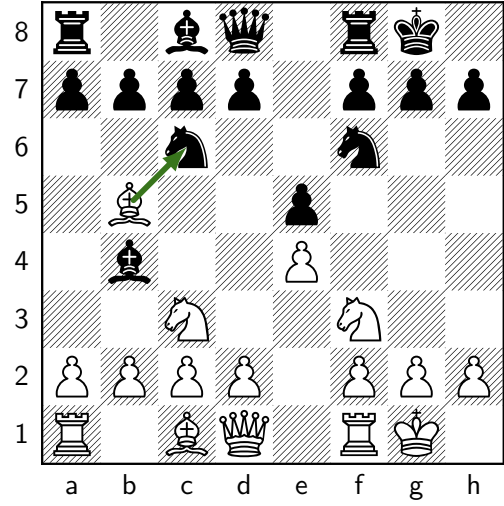
Melanie (Deep_Vision)

Abstract

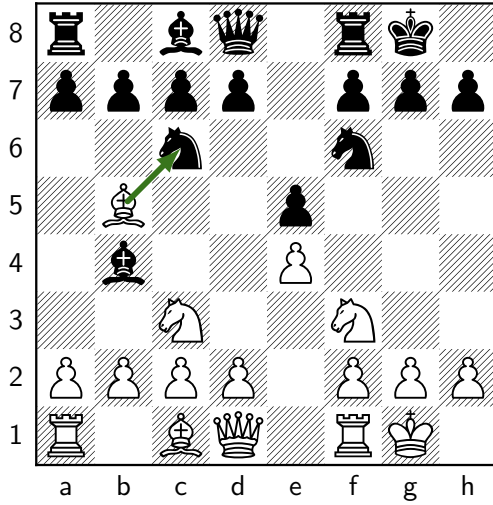
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie's move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen's role) are called out, as no material/eval engine explains a move that way.



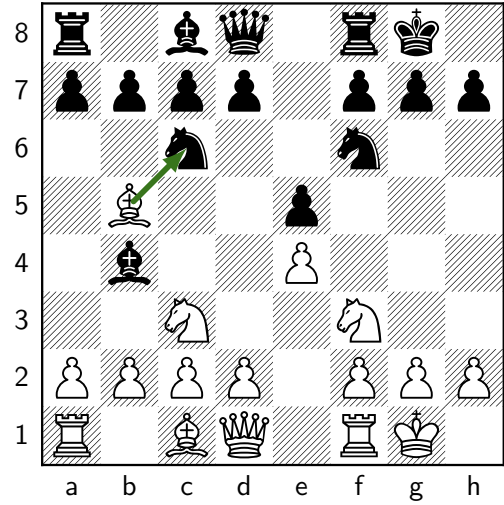
(a)



(b)

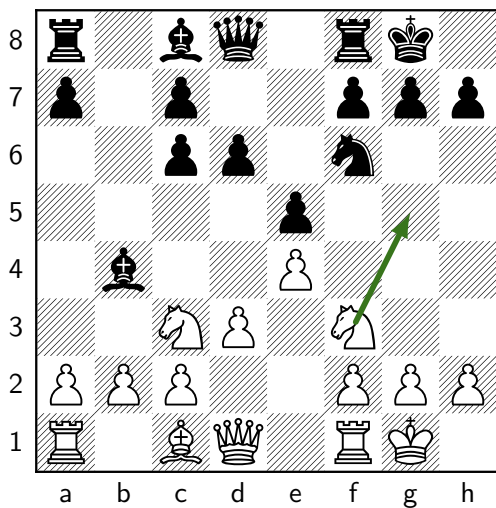


(c)

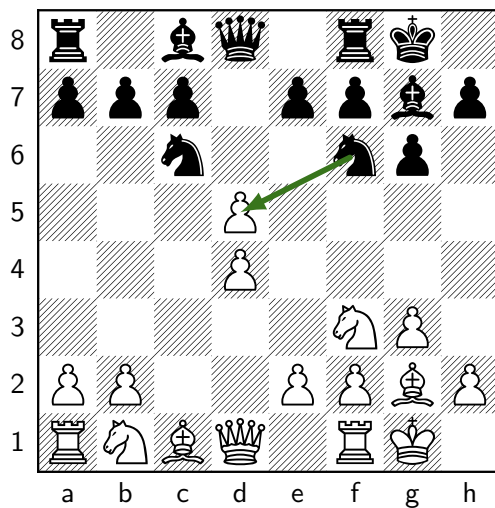


(d)

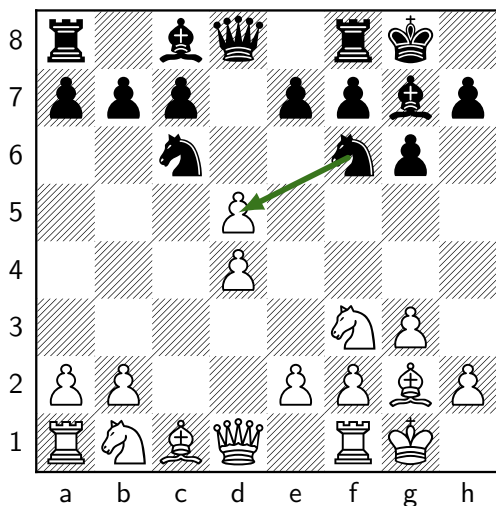
Figure 1: **Move 7, White: Bxc6**. (a) Our position; Melanie plays Bxc6 (b5c6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#1360` (ply 10) [structural distance $d=37$, QAP obj=63]: the master played Bxc6; under μ our Bb5 plays the role of that game’s Bb5; cross-type roles: our Pa2 as their Ra1, our Pd3 as their Bc1; transferred move b5c6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#1361` (ply 10) [structural distance $d=37$, QAP obj=63]: the master played Bxc6; under μ our Bb5 plays the role of that game’s Bb5; cross-type roles: our Pa2 as their Ra1, our Pd3 as their Bc1; transferred move b5c6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#164` (ply 10) [structural distance $d=37$, QAP obj=63]: the master played Bxc6; under μ our Bb5 plays the role of that game’s Bb5; cross-type roles: our Pa2 as their Ra1, our Pd3 as their Bc1; transferred move b5c6 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Bxc6 (summed QAP objective 254, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



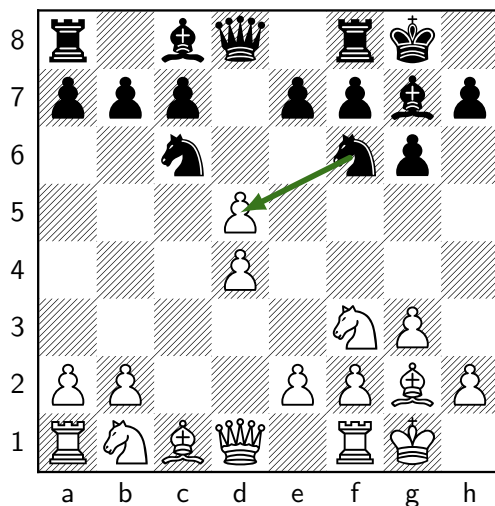
(a)



(b)

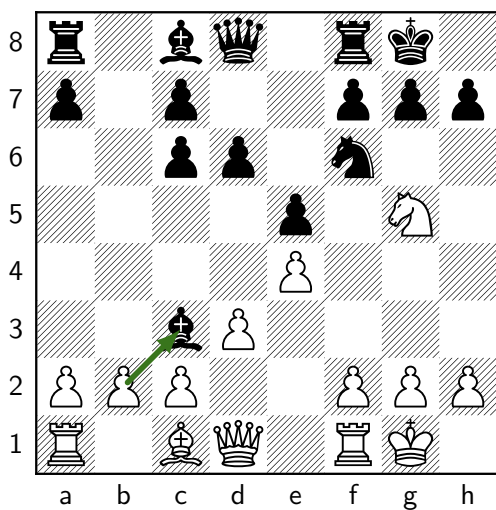


(c)

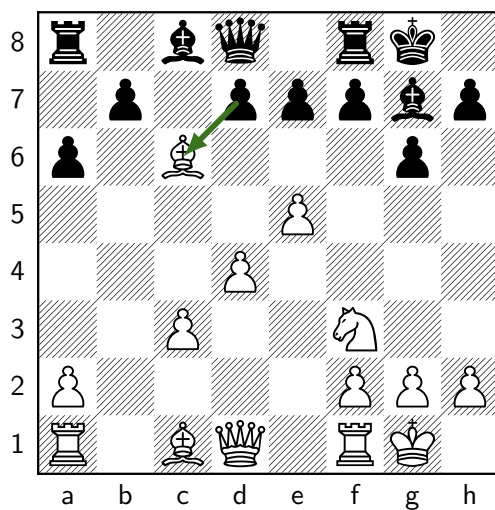


(d)

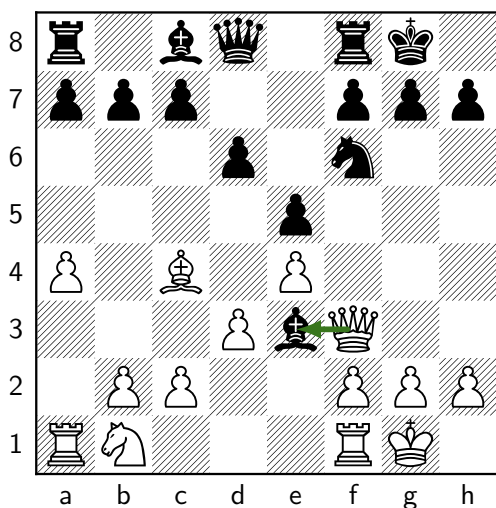
Figure 2: **Move 8, White: Ng5.** (a) Our position; Melanie plays Ng5 (f3g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2171` (ply 13) [structural distance $d=57$, QAP obj=27]: the master played Nxd5; under μ our Nf3 plays the role of that game’s Nf3; cross-type roles: our pe5 as their Ra8, our nf6 as their Bc8; transferred move f3g5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2172` (ply 13) [structural distance $d=57$, QAP obj=27]: the master played Nxd5; under μ our Nf3 plays the role of that game’s Nf3; cross-type roles: our pe5 as their Ra8, our nf6 as their Bc8; transferred move f3g5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2884` (ply 13) [structural distance $d=57$, QAP obj=27]: the master played Nxd5; under μ our Nf3 plays the role of that game’s Nf3; cross-type roles: our pe5 as their Ra8, our nf6 as their Bc8; transferred move f3g5 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Ng5 (summed QAP objective 171, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



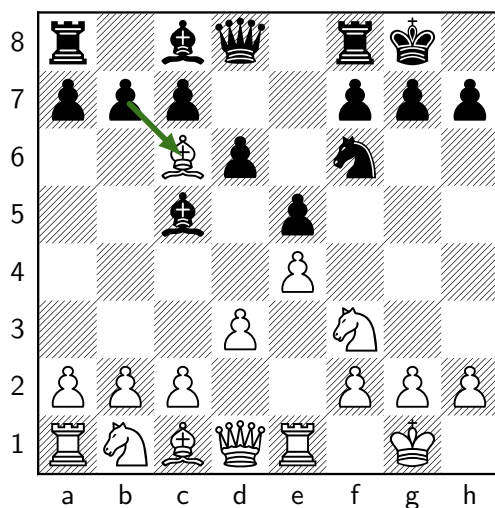
(a)



(b)

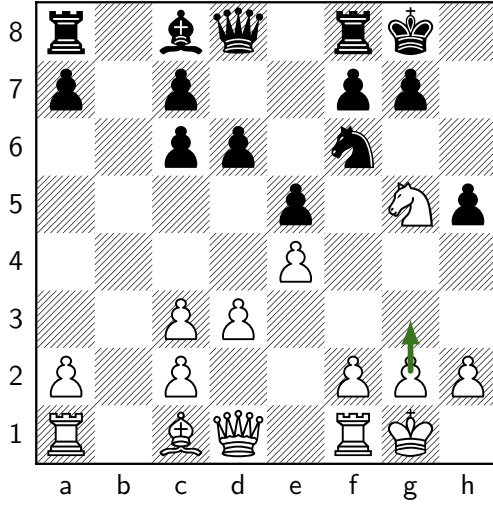


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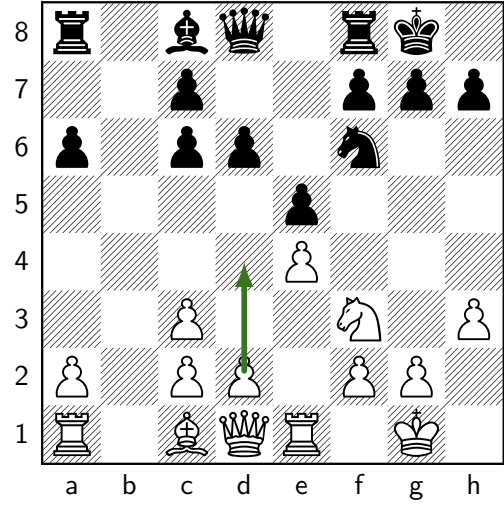


(d)

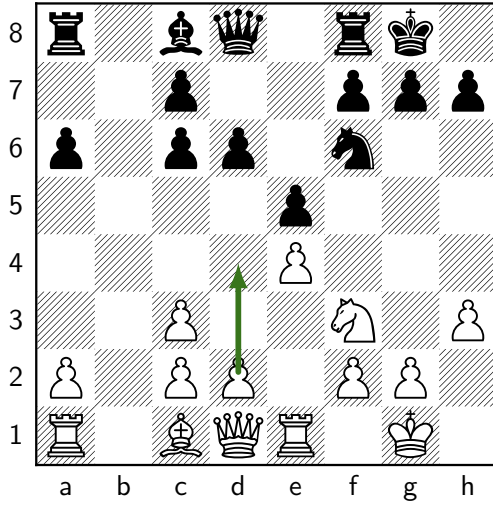
Figure 3: **Move 9, White: bxc3.** (a) Our position; Melanie plays **bxc3** (**b2c3**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1295` (ply 21) [structural distance $d=60$, QAP obj=83]: the master played **dx6**; under μ our *Pb2* plays the role of that game’s *Pd2*; cross-type roles: our *Nf6* as their *Pe7*, our *Ng5* as their *Bg7*; transferred move **b2c3** (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#668` (ply 18) [structural distance $d=61$, QAP obj=69]: the master played **Qxe3**; under μ our *Pb2* plays the role of that game’s *Qf3* (a **cross-type** correspondence); cross-type roles: our *Pc2* as their *Nb1*, our *Bc1* as their *Pb2*; transferred move **b2c3** (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#2359` (ply 13) [structural distance $d=68$, QAP obj=69]: the master played **bxc6**; under μ our *Pb2* plays the role of that game’s *Pb2*; cross-type roles: our *Pc2* as their *Nf6*, our *Ng5* as their *Bc5*; transferred move **b2c3** (**endorses this move**). Synthesis: 32 retrieved analogs transferred to **bxc3** (summed QAP objective 1471, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



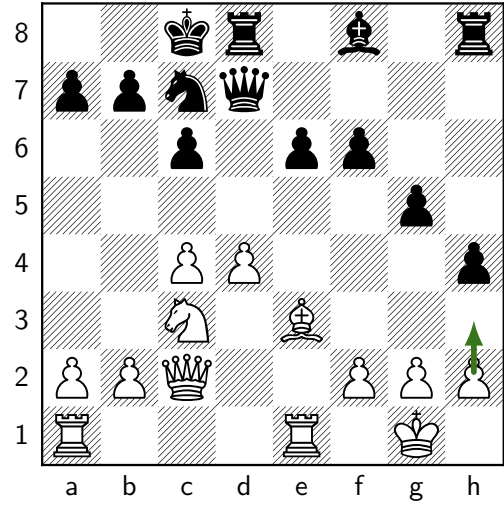
(a)



(b)

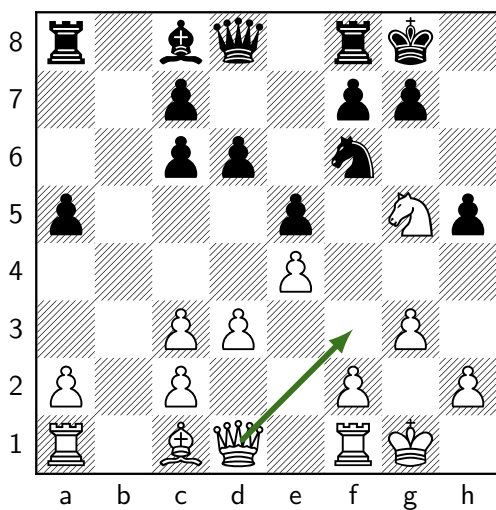


(c)

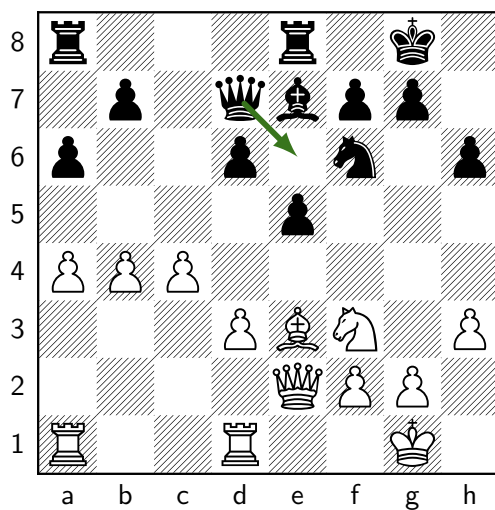


(d)

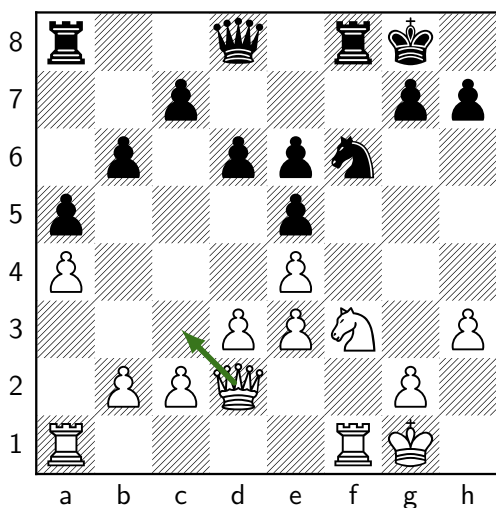
Figure 4: **Move 10, White: g3**. (a) Our position; Melanie plays **g3** (**g2g3**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1503` (ply 18) [structural distance $d=52$, QAP obj=52]: the master played **d4**; under μ our $Pg2$ plays the role of that game’s $Pd2$; cross-type roles: our $Pc3$ as their $Ra1$, our $Pc2$ as their $Bc1$; transferred move **g2g3** (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#1504` (ply 18) [structural distance $d=52$, QAP obj=52]: the master played **d4**; under μ our $Pg2$ plays the role of that game’s $Pd2$; cross-type roles: our $Pc3$ as their $Ra1$, our $Pc2$ as their $Bc1$; transferred move **g2g3** (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#3988` (ply 30) [structural distance $d=60$, QAP obj=40]: the master played **h3**; under μ our $Pg2$ plays the role of that game’s $Ph2$; cross-type roles: our $Qd1$ as their $Ra1$, our $Ra1$ as their $Pa2$; transferred move **g2g3** (**endorses this move**). Synthesis: 6 retrieved analogs transferred to **g3** (summed QAP objective 215, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



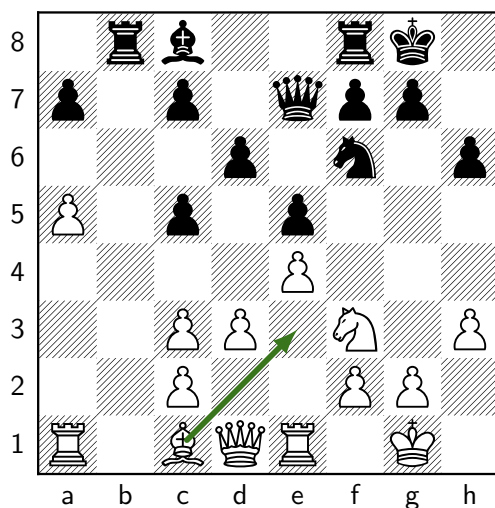
(a)



(b)

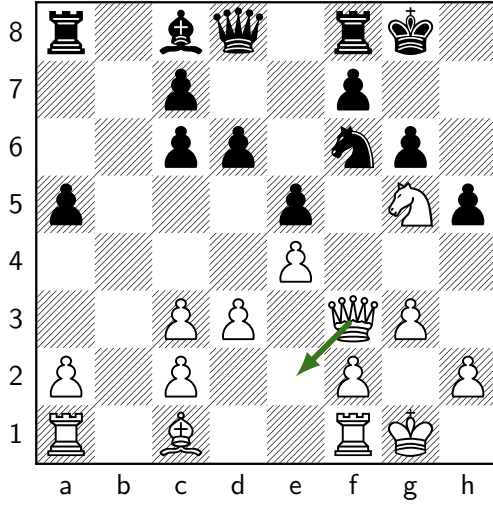


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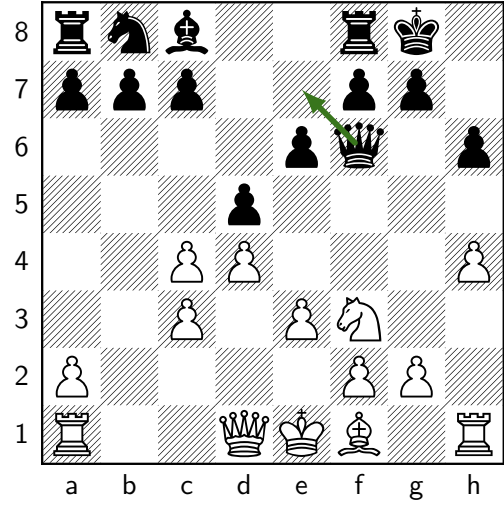


(d)

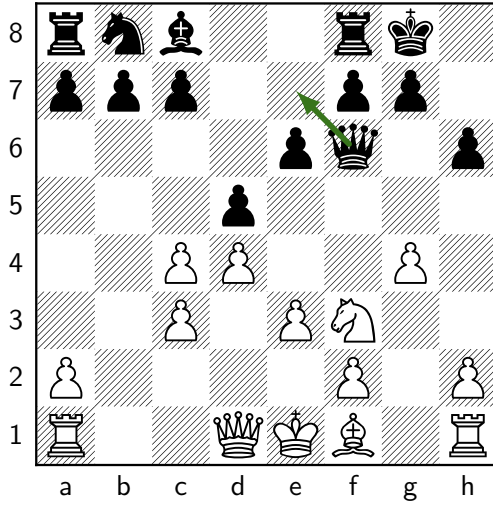
Figure 5: **Move 11, White: Qf3.** (a) Our position; Melanie plays Qf3 (d1f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#3257` (ply 35) [structural distance $d=63$, QAP obj=59]: the master played Qe6; under μ our Qd1 plays the role of that game's Qd2; cross-type roles: our Bc1 as their Pb7, our pe5 as their Be7; transferred move d1f3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#1451` (ply 28) [structural distance $d=61$, QAP obj=54]: the master played Qc3; under μ our Qd1 plays the role of that game's Qd2; cross-type roles: our pa5 as their Ra1, our Ra1 as their Pb2; transferred move d1f3 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#40` (ply 24) [structural distance $d=61$, QAP obj=47]: the master played Be3; under μ our Qd1 plays the role of that game's Bc1 (a **cross-type** correspondence); cross-type roles: our bc8 as their Qd1, our Pf2 as their Re1; transferred move d1f3 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qf3 (summed QAP objective 202, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



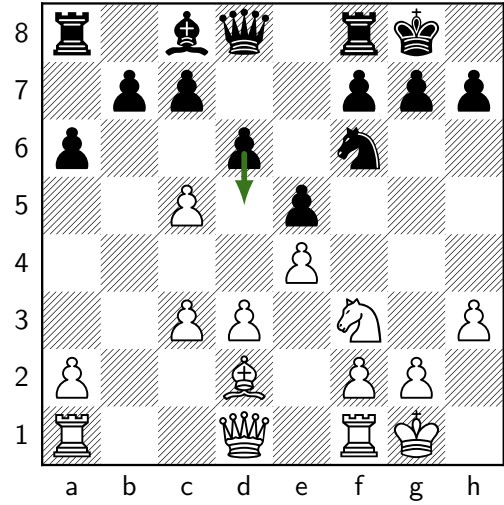
(a)



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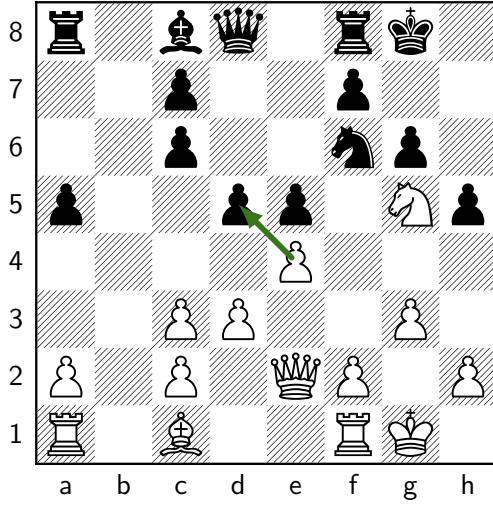


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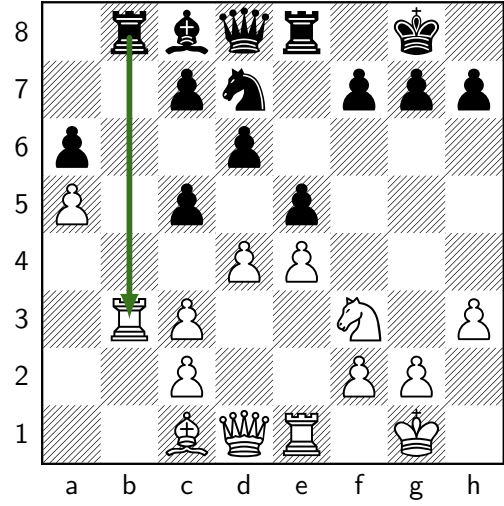


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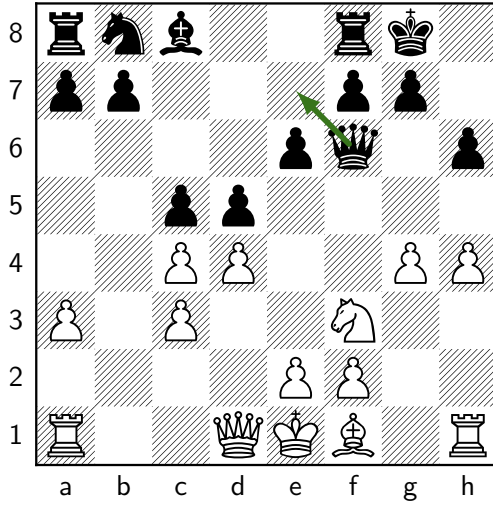
Figure 6: **Move 12, White: Qe2.** (a) Our position; Melanie plays Qe2 (f3e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#773` (ply 17) [structural distance $d=76$, QAP obj=63]: the master played Qe7; under μ our Qf3 plays the role of that game's Qf3; cross-type roles: our Ph2 as their Ra8, our pd6 as their Nb8; transferred move f3e2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#772` (ply 17) [structural distance $d=60$, QAP obj=56]: the master played Qe7; under μ our Qf3 plays the role of that game's Qf3; cross-type roles: our Ra1 as their Nb8, our pc7 as their Bc8; transferred move f3e2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3346` (ply 21) [structural distance $d=73$, QAP obj=38]: the master played d5; under μ our Qf3 plays the role of that game's Pd3 (a **cross-type** correspondence); cross-type roles: our pc6 as their Ra8, our Pe4 as their Bc8; transferred move f3e2 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Qe2 (summed QAP objective 259, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



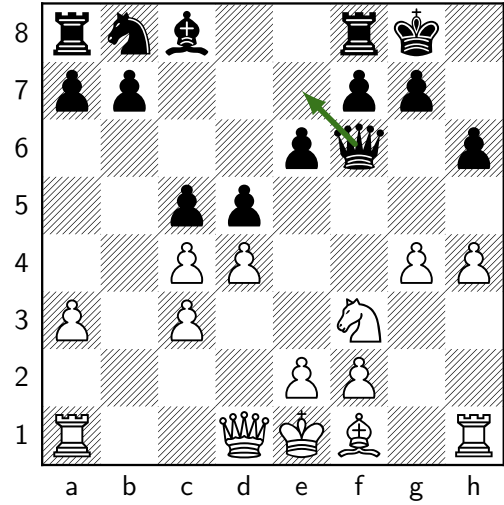
(a)



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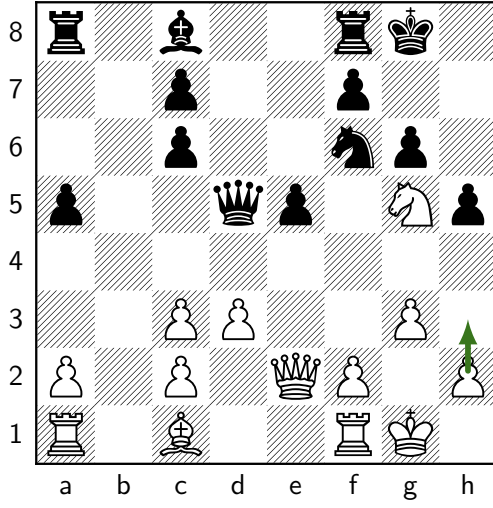


(c)

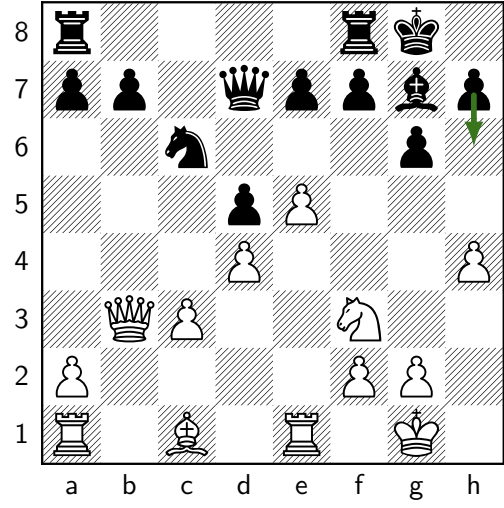


(d)

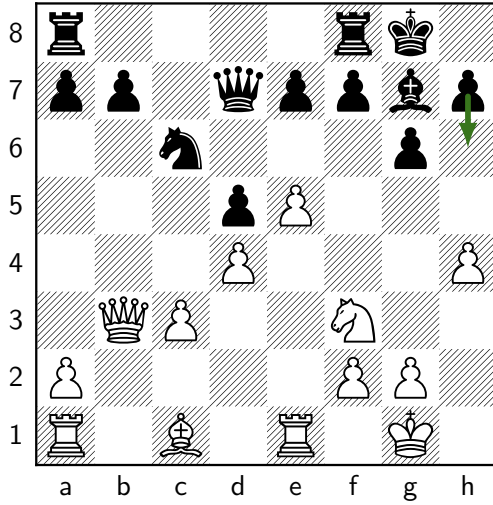
Figure 7: **Move 13, White: exd5.** (a) Our position; Melanie plays exd5 (e4d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#1503 (ply 27) [structural distance $d=65$, QAP obj=123]: the master played Rxb3; under μ our Pe_4 plays the role of that game’s $Rb1$ (a **cross-type** correspondence); cross-type roles: our Pc_2 as their $Bc8$, our $Ra1$ as their $Pc7$; transferred move e4d5 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#2315 (ply 19) [structural distance $d=76$, QAP obj=101]: the master played Qe7; under μ our Pe_4 plays the role of that game’s $Qf3$ (a **cross-type** correspondence); cross-type roles: our pe_5 as their $Ra8$, our $Ra1$ as their $Nb8$; transferred move e4d5 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#2316 (ply 19) [structural distance $d=76$, QAP obj=101]: the master played Qe7; under μ our Pe_4 plays the role of that game’s $Qf3$ (a **cross-type** correspondence); cross-type roles: our pe_5 as their $Ra8$, our $Ra1$ as their $Nb8$; transferred move e4d5 (**endorses this move**). Synthesis: 11 retrieved analogs transferred to exd5 (summed QAP objective 768, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



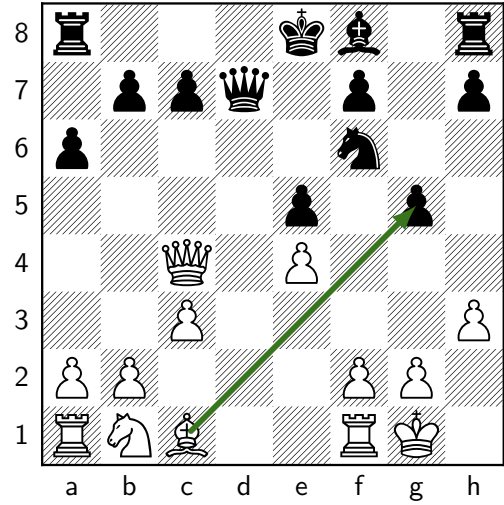
(a)



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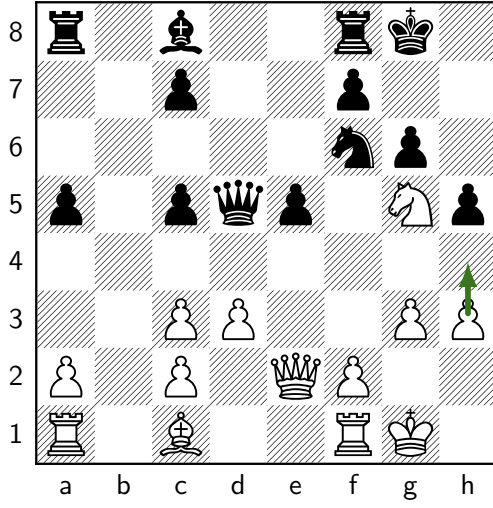


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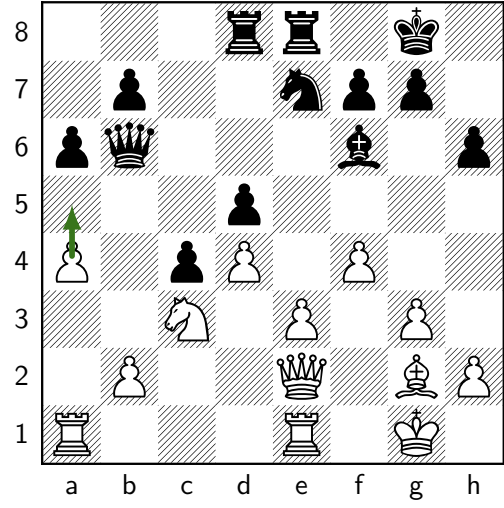


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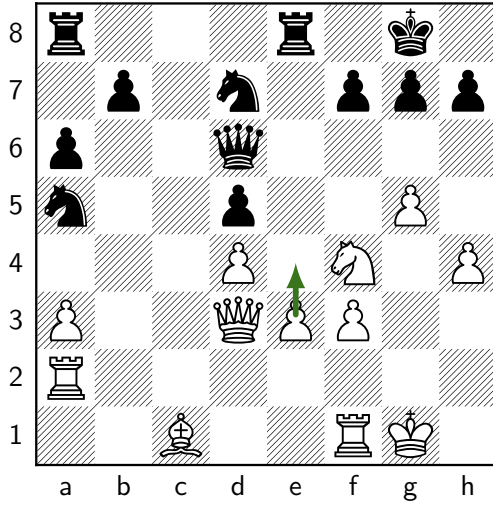
Figure 8: **Move 14, White: h3.** (a) Our position; Melanie plays **h3** (h2h3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2693` (ply 29) [structural distance $d=72$, QAP obj=88]: the master played **h6**; under μ our *Ph2* plays the role of that game’s *Ph2*; cross-type roles: our *Pc2* as their *Ra8*, our *Bc1* as their *Qd7*; transferred move **h2h3** (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1975` (ply 29) [structural distance $d=72$, QAP obj=88]: the master played **h6**; under μ our *Ph2* plays the role of that game’s *Ph2*; cross-type roles: our *Pc2* as their *Ra8*, our *Bc1* as their *Qd7*; transferred move **h2h3** (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#737` (ply 24) [structural distance $d=73$, QAP obj=90]: the master played **Bxg5**; under μ our *Qe2* plays the role of that game’s *Bc1* (a **cross-type** correspondence); cross-type roles: our *Bc1* as their *Ra1*, our *pg6* as their *Nb1*; transferred move **e2e5**. Synthesis: 2 retrieved analogs transferred to **h3** (summed QAP objective 176, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



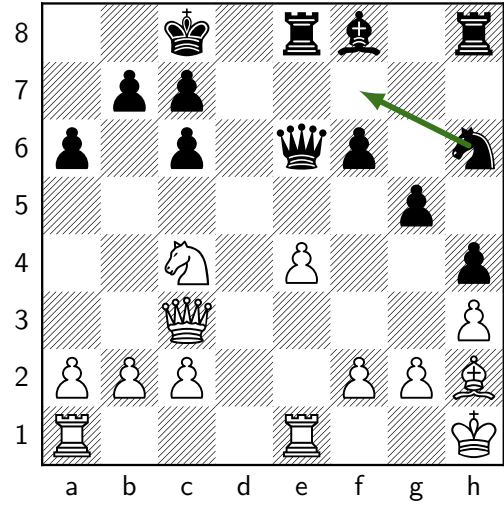
(a)



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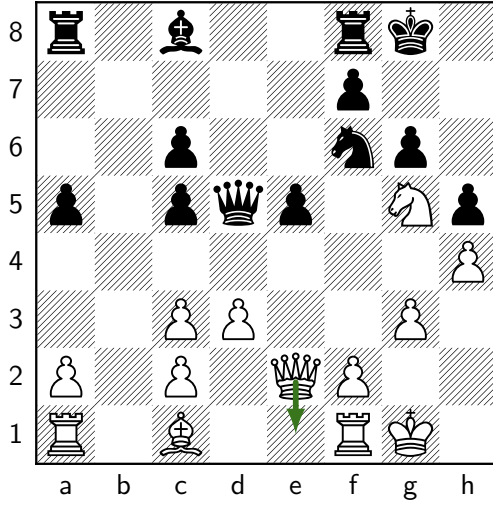


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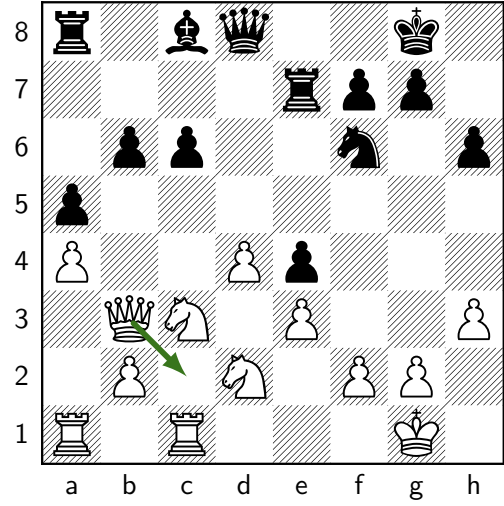


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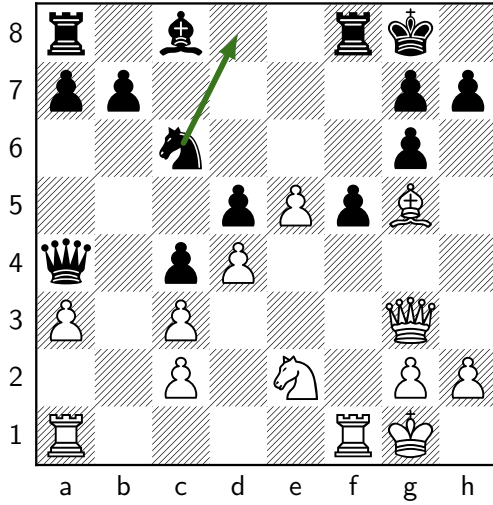
Figure 9: **Move 15, White: h4**. (a) Our position; Melanie plays **h4** (**h3h4**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#1124` (ply 36) [structural distance $d=74$, QAP obj=93]: the master played **a5**; under μ our *Ph3* plays the role of that game’s *Pa4*; cross-type roles: our *Pc2* as their *Ra1*, our *Ng5* as their *Bg2*; transferred move **h3h4** (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#2931` (ply 36) [structural distance $d=71$, QAP obj=25]: the master played **e4**; under μ our *Ph3* plays the role of that game’s *Pe3*; cross-type roles: our *Ra1* as their *Bc1*, our *Pc2* as their *Ra2*; transferred move **h3h4** (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#87` (ply 37) [structural distance $d=76$, QAP obj=24]: the master played **Nf7**; under μ our *Ph3* plays the role of that game’s *Nh3* (a **cross-type** correspondence); cross-type roles: our *pc7* as their *Bf8*, our *Bc1* as their *Rh8*; transferred move **h3h4** (**endorses this move**). Synthesis: 4 retrieved analogs transferred to **h4** (summed QAP objective 163, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



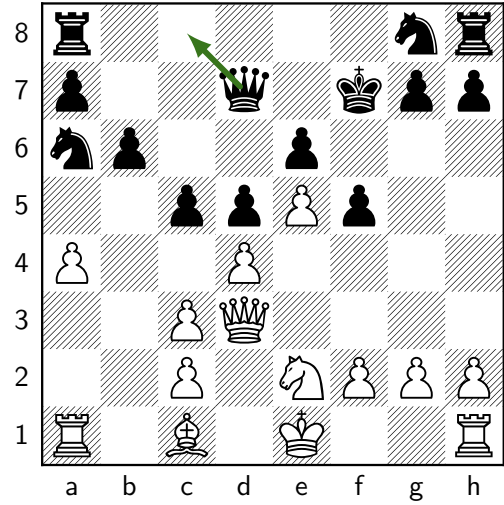
(a)



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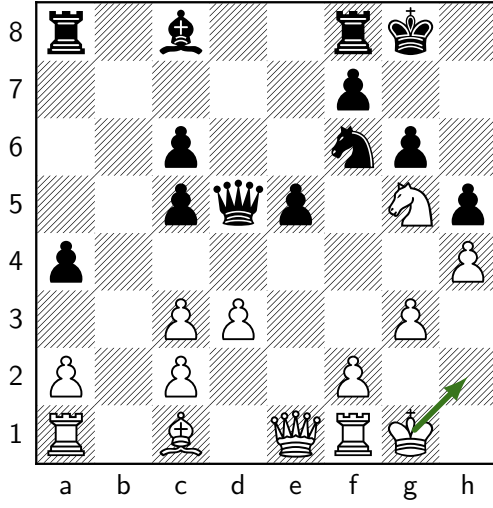


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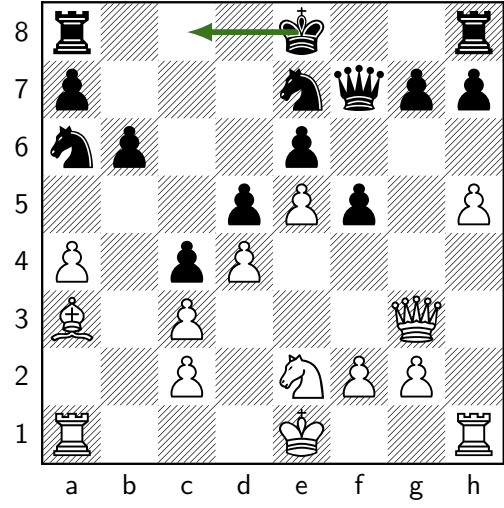


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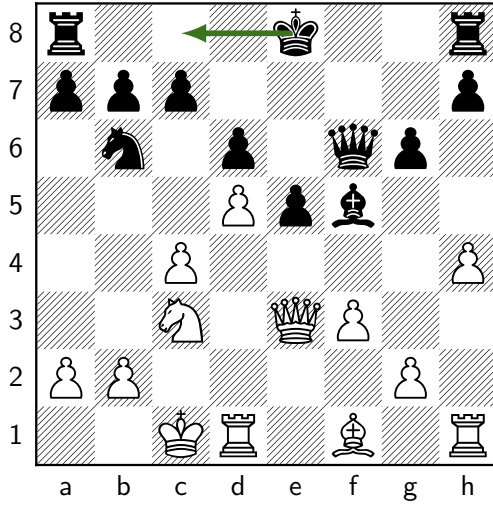
Figure 10: **Move 16, White: Qe1**. (a) Our position; Melanie plays Qe1 (e2e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#238` (ply 34) [structural distance $d=83$, QAP obj=78]: the master played Qc2; under μ our Qe2 plays the role of that game’s Qb3; cross-type roles: our Pc3 as their Ra1, our Bc1 as their Rc1; transferred move e2e1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2883` (ply 29) [structural distance $d=84$, QAP obj=49]: the master played Nd8; under μ our Qe2 plays the role of that game’s Nc3 (a **cross-type** correspondence); cross-type roles: our Pf2 as their Ra8, our Ra8 as their Bc8; transferred move e2e1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#517` (ply 23) [structural distance $d=83$, QAP obj=34]: the master played Qc8; under μ our Qe2 plays the role of that game’s Qd2; cross-type roles: our ph5 as their Ra8, our nf6 as their Pg7; transferred move e2e1 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Qe1 (summed QAP objective 225, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



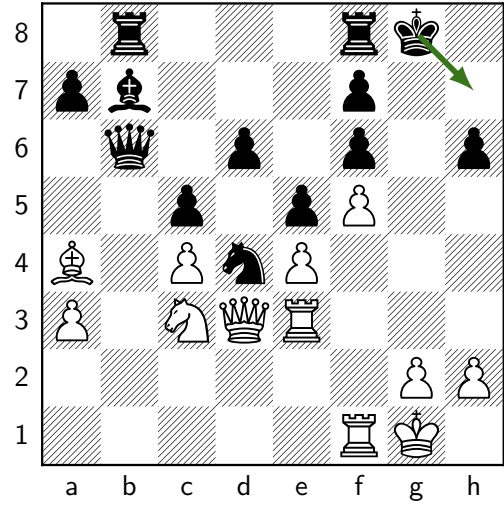
(a)



(b)

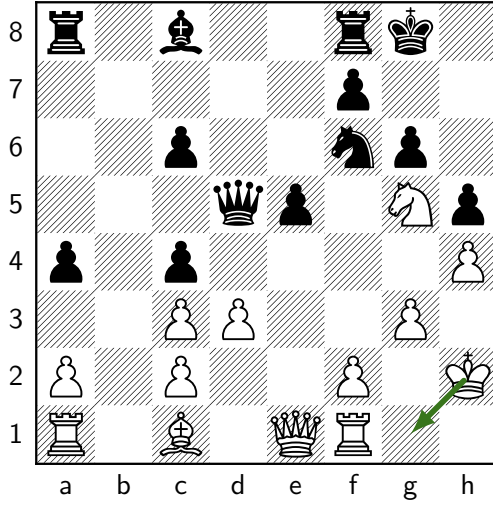


(c)

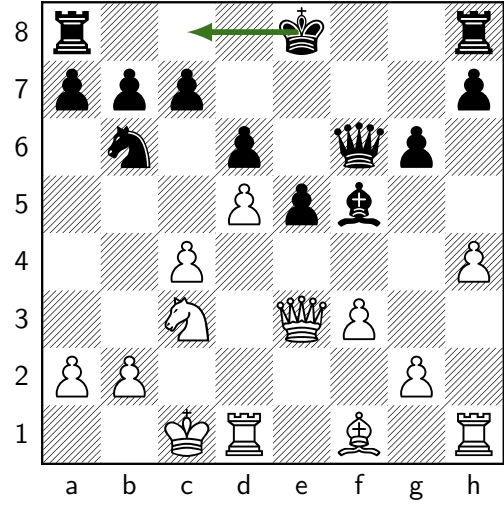


(d)

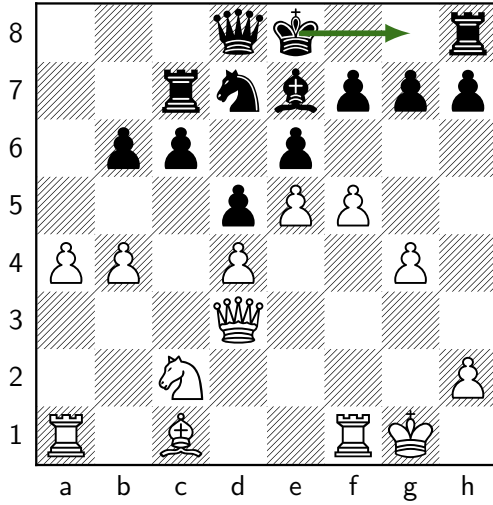
Figure 11: **Move 17, White: Kh2.** (a) Our position; Melanie plays Kh2 (**g1h2**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#518` (ply 27) [structural distance $d=85$, QAP obj=37]: the master played 0-0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Pg3* as their *Rh8*, our *pf7* as their *Ne7*; transferred move **g1h2 (endorses this move)**. (c) Retrieved from `ccr1_top_p0002#909` (ply 31) [structural distance $d=82$, QAP obj=36]: the master played 0-0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *pc6* as their *Rh8*, our *Bc1* as their *Pb7*; transferred move **g1h2 (endorses this move)**. (d) Retrieved from `ccr1_top_p0000#1340` (ply 45) [structural distance $d=85$, QAP obj=8]: the master played Kh7; under μ our *Kg1* plays the role of that game’s *Kg1*; cross-type roles: our *pc6* as their *Rb8*, our *Ra1* as their *Pa7*; transferred move **g1h2 (endorses this move)**. Synthesis: 3 retrieved analogs transferred to Kh2 (summed QAP objective 81, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



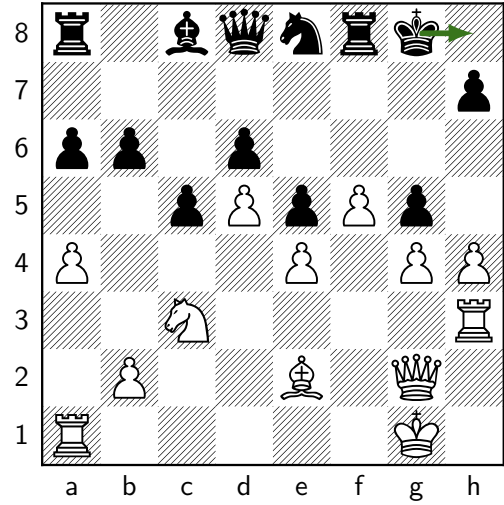
(a)



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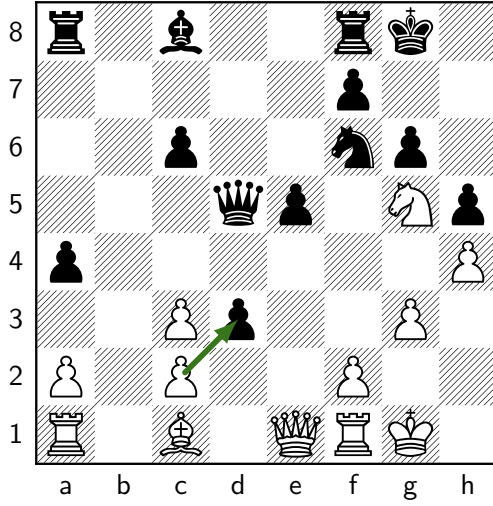


(c)

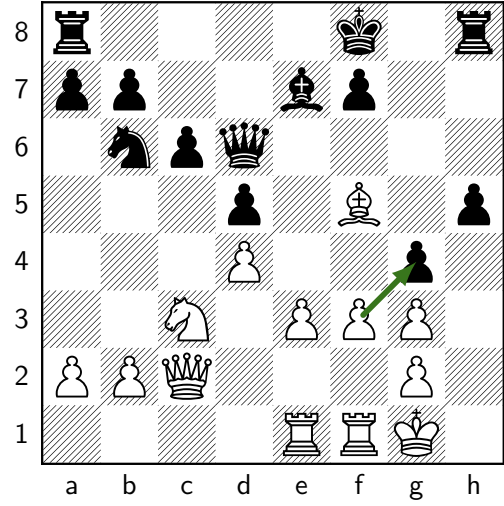


(d)

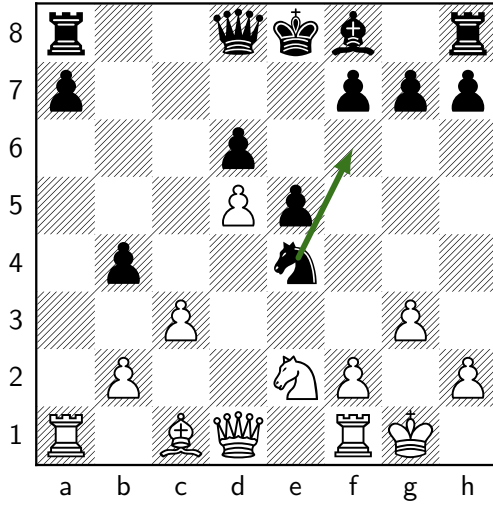
Figure 12: **Move 18, White: Kg1**. (a) Our position; Melanie plays Kg1 (h2g1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#909` (ply 31) [structural distance $d=89$, QAP obj=61]: the master played 0-0-0; under μ our *Kh2* plays the role of that game's *Ke1*; cross-type roles: our *pa4* as their *Rh8*, our *Pd3* as their *Nb6*; transferred move h2g1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3438` (ply 37) [structural distance $d=88$, QAP obj=56]: the master played 0-0; under μ our *Kh2* plays the role of that game's *Ke1*; cross-type roles: our *Rf1* as their *Qd8*, our *Pc3* as their *Rc7*; transferred move h2g1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1309` (ply 47) [structural distance $d=77$, QAP obj=54]: the master played Kh8; under μ our *Kh2* plays the role of that game's *Kg1*; cross-type roles: our *qd5* as their *Ra8*, our *Ng5* as their *Bc8*; transferred move h2g1 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kg1 (summed QAP objective 196, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



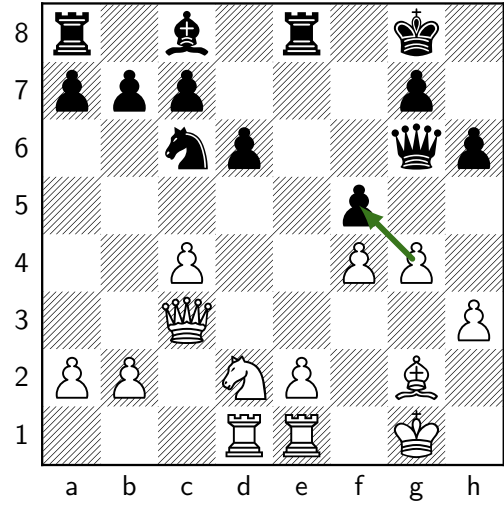
(a)



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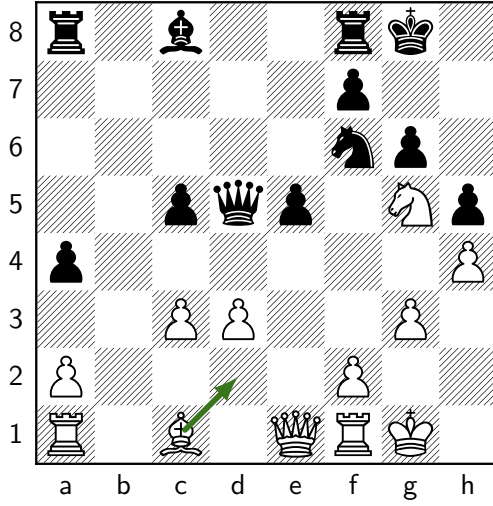


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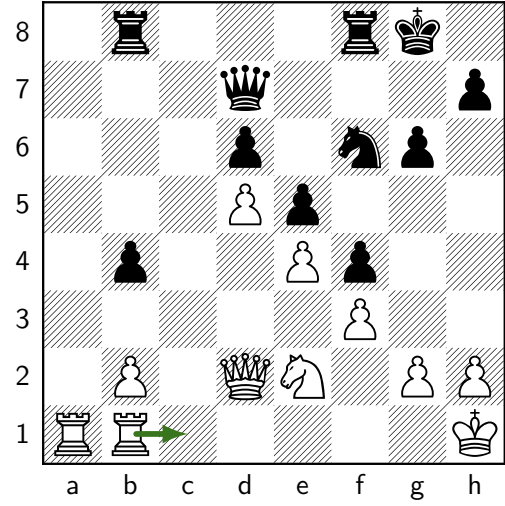


(d)

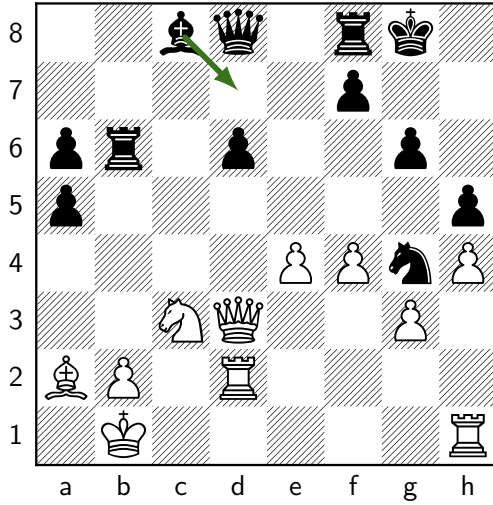
Figure 13: **Move 19, White: cxd3.** (a) Our position; Melanie plays cxd3 (c2d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#269` (ply 36) [structural distance $d=42$, QAP obj=72]: the master played **fxg4**; under μ our *Pc2* plays the role of that game’s *Pf3*; cross-type roles: our *Bc1* as their *Rf1*, our *Ra1* as their *Pa2*; transferred move **c2d3** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1863` (ply 27) [structural distance $d=73$, QAP obj=60]: the master played **Nf6**; under μ our *Pc2* plays the role of that game’s *Ne5* (a **cross-type** correspondence); cross-type roles: our *Qe1* as their *Ra8*, our *Rf1* as their *Qd8*; transferred move **c2d3** (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1094` (ply 36) [structural distance $d=74$, QAP obj=58]: the master played **gxf5**; under μ our *Pc2* plays the role of that game’s *Pg4*; cross-type roles: our *Qe1* as their *Rd1*, our *Ra1* as their *Pa2*; transferred move **c2d3** (**endorses this move**). Synthesis: 12 retrieved analogs transferred to **cxd3** (summed QAP objective 424, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



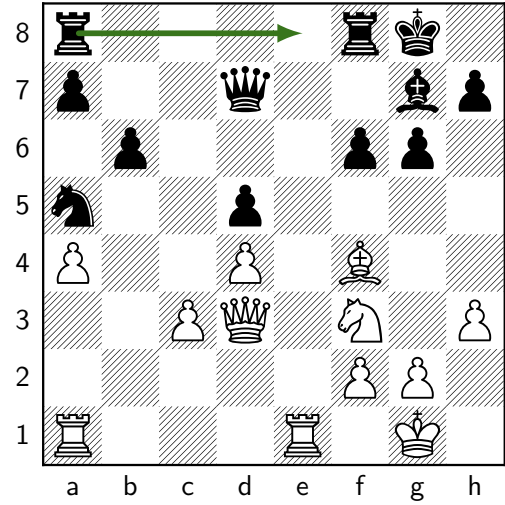
(a)



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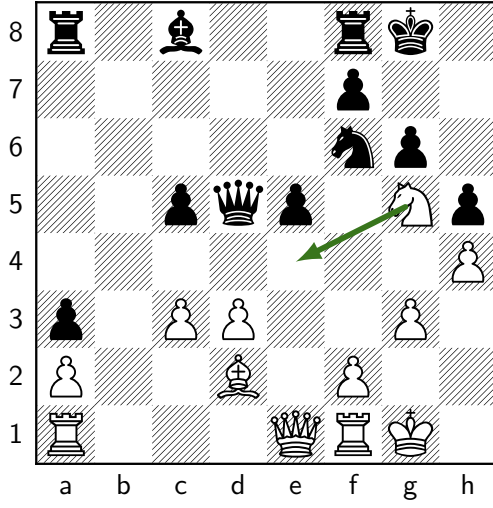


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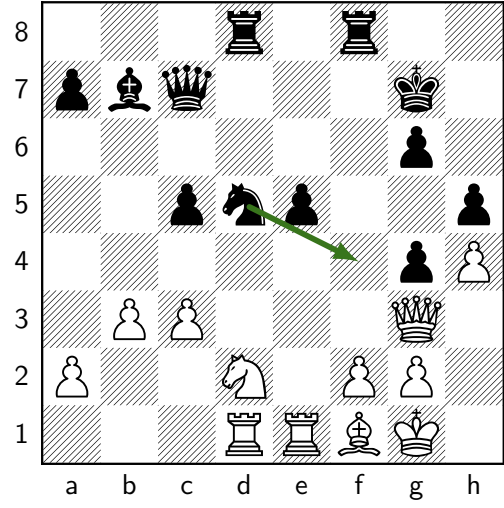


(d)

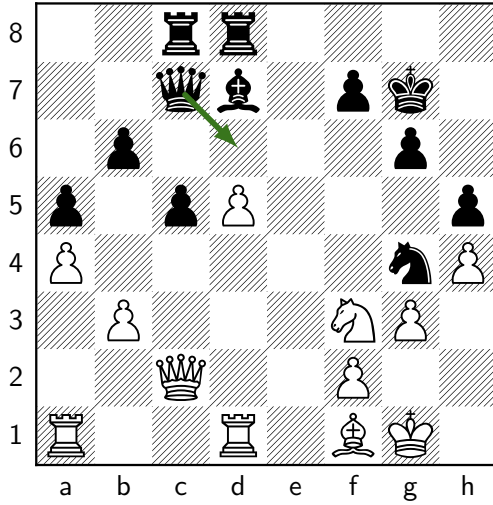
Figure 14: **Move 20, White: Bd2.** (a) Our position; Melanie plays Bd2 (c1d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#689` (ply 46) [structural distance $d=75$, QAP obj=88]: the master played Rc1; under μ our Bc1 plays the role of that game’s Rb1 (a **cross-type** correspondence); cross-type roles: our Ng5 as their Qd2, our Rf1 as their Pg2; transferred move c1d2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#563` (ply 53) [structural distance $d=65$, QAP obj=59]: the master played Bd7; under μ our Bc1 plays the role of that game’s Bc1; cross-type roles: our Ng5 as their Rb6, our nf6 as their Pa5; transferred move c1d2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#492` (ply 37) [structural distance $d=73$, QAP obj=43]: the master played Rae8; under μ our Bc1 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our ra8 as their Bg7, our Qe1 as their Pf6; transferred move c1d2 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Bd2 (summed QAP objective 214, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



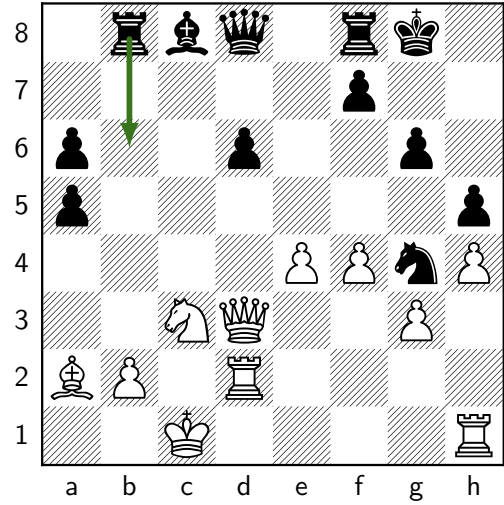
(a)



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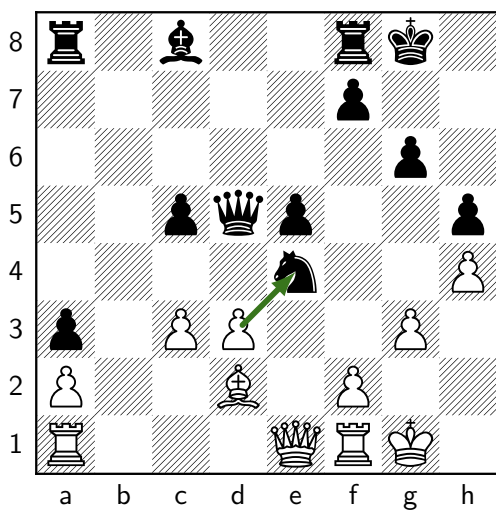


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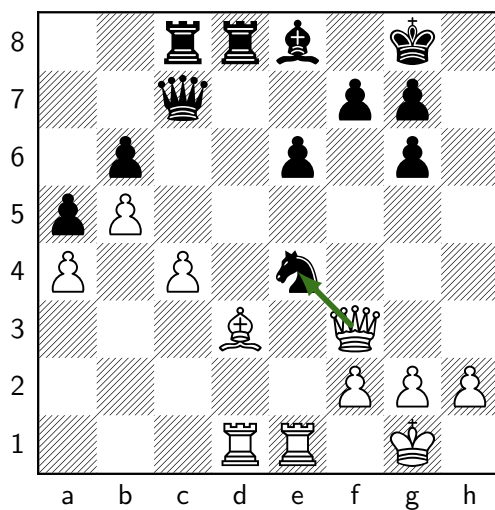


(d)

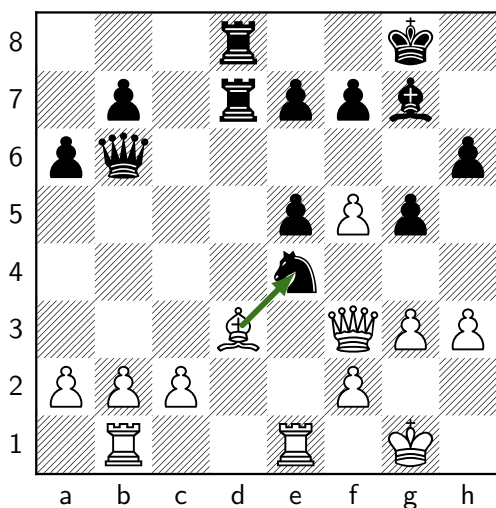
Figure 15: **Move 21, White: Ne4.** (a) Our position; Melanie plays Ne4 (g5e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1034` (ply 53) [structural distance $d=77$, QAP obj=67]: the master played Nf4; under μ our Ng5 plays the role of that game’s Nd4; cross-type roles: our Bd2 as their Rd8, our Pg3 as their Rf8; transferred move g5e4 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#634` (ply 51) [structural distance $d=62$, QAP obj=92]: the master played Qd6; under μ our Qe1 plays the role of that game’s Qc2; cross-type roles: our Bd2 as their Rd8, our qd5 as their Bd7; transferred move e1e2. (d) Retrieved from `ccrl_top_p0002#563` (ply 51) [structural distance $d=61$, QAP obj=89]: the master played Rb6; under μ our Ng5 plays the role of that game’s Rb1 (a **cross-type** correspondence); cross-type roles: our Ra1 as their Bc8, our pg6 as their Ng4; transferred move g5h3. Synthesis: 2 retrieved analogs transferred to Ne4 (summed QAP objective 134, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



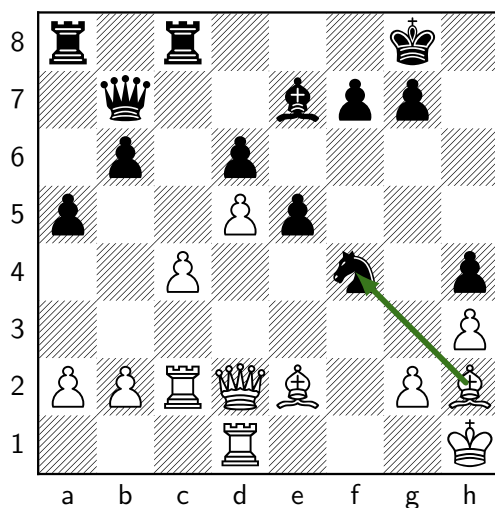
(a)



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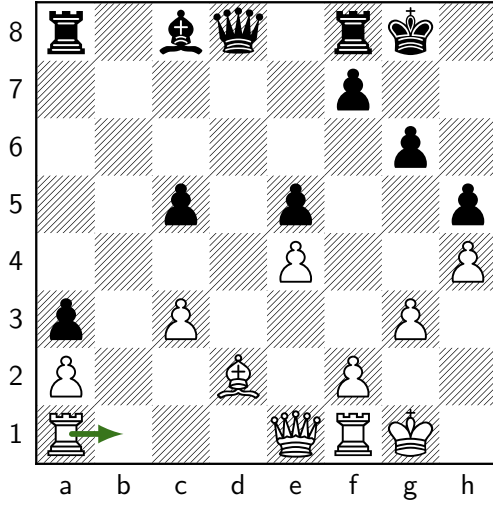


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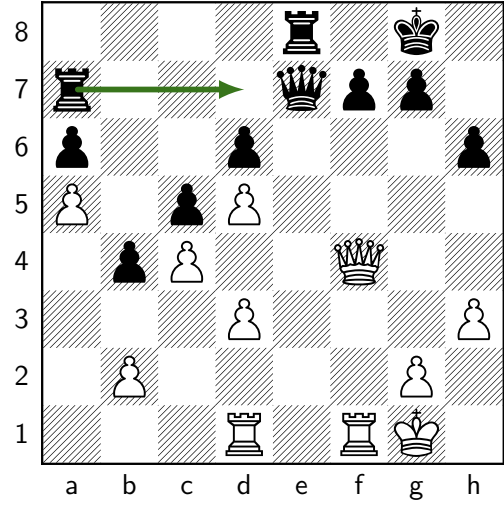


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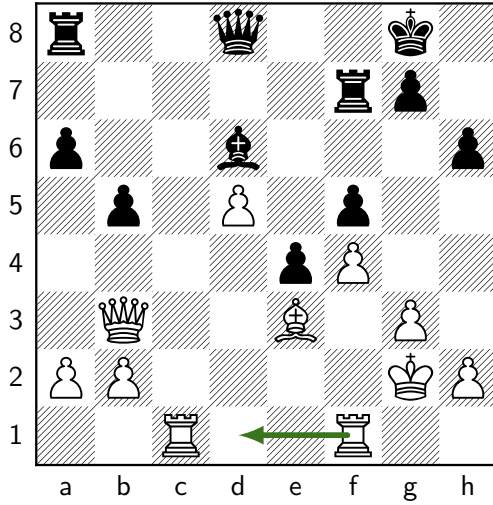
Figure 16: **Move 22, White: dxe4.** (a) Our position; Melanie plays dxe4 (d3e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1221` (ply 44) [structural distance $d=90$, QAP obj=86]: the master played **Qxe4**; under μ our *Pd3* plays the role of that game's *Qf3* (a **cross-type** correspondence); cross-type roles: our *Qe1* as their *Re1*, our *Bd2* as their *Pf2*; transferred move **d3e4 (endorses this move)**. (c) Retrieved from `ccr1_top_p0003#2324` (ply 42) [structural distance $d=93$, QAP obj=79]: the master played **Bxe4**; under μ our *Pd3* plays the role of that game's *Bd3* (a **cross-type** correspondence); cross-type roles: our *Qe1* as their *Re1*, our *Rf1* as their *Pc2*; transferred move **d3e4 (endorses this move)**. (d) Retrieved from `ccr1_top_p0002#195` (ply 48) [structural distance $d=68$, QAP obj=77]: the master played **Bxf4**; under μ our *Pd3* plays the role of that game's *Bh2* (a **cross-type** correspondence); cross-type roles: our *bc8* as their *Rc2*, our *Bd2* as their *Ph3*; transferred move **d3e4 (endorses this move)**. Synthesis: 16 retrieved analogs transferred to dxe4 (summed QAP objective 838, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



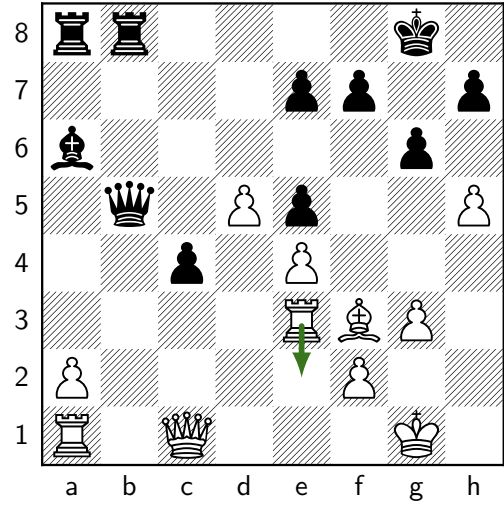
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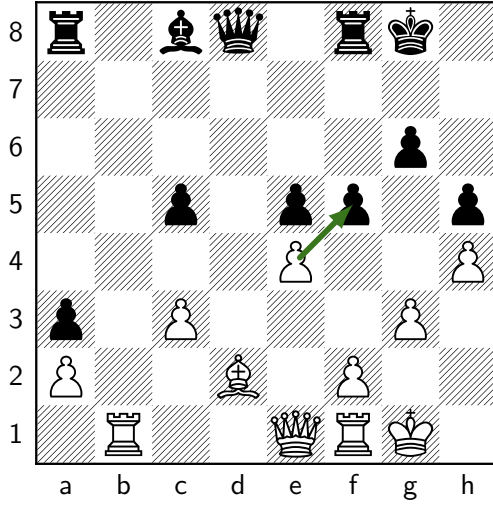


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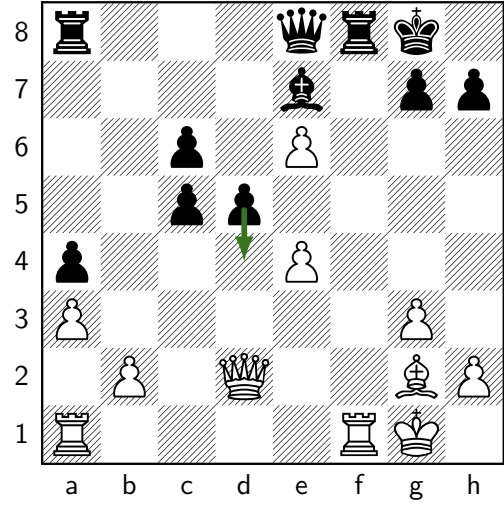


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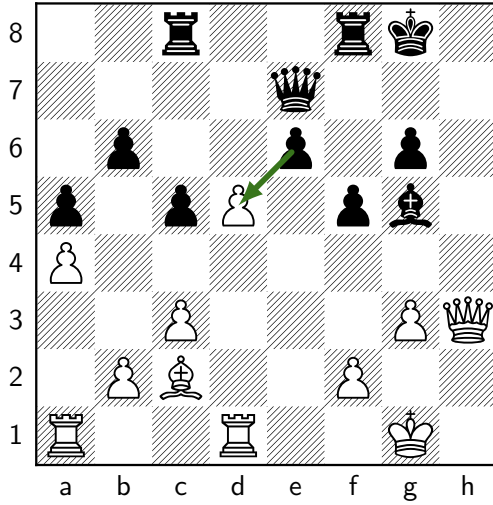
Figure 17: **Move 23, White: Rb1.** (a) Our position; Melanie plays Rb1 (a1b1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2918` (ply 51) [structural distance $d=62$, QAP obj=56]: the master played Rb7; under μ our *Ra1* plays the role of that game’s *Ra2*; cross-type roles: our *Pg3* as their *Qe7*, our *Bd2* as their *Pd6*; transferred move a1b1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3279` (ply 42) [structural distance $d=72$, QAP obj=33]: the master played Rf1; under μ our *Ra1* plays the role of that game’s *Rf1*; cross-type roles: our *Pg3* as their *Rc1*, our *Rf1* as their *Pb2*; transferred move a1b1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1031` (ply 56) [structural distance $d=57$, QAP obj=14]: the master played Re2; under μ our *Ra1* plays the role of that game’s *Re3*; cross-type roles: our *pc5* as their *Ra1*, our *Rf1* as their *Qc1*; transferred move a1b1 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Rb1 (summed QAP objective 108, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



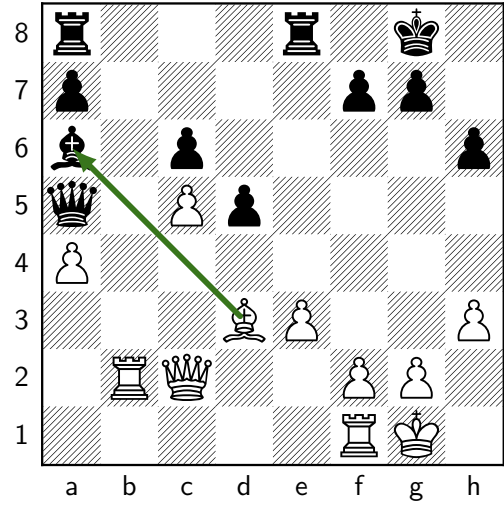
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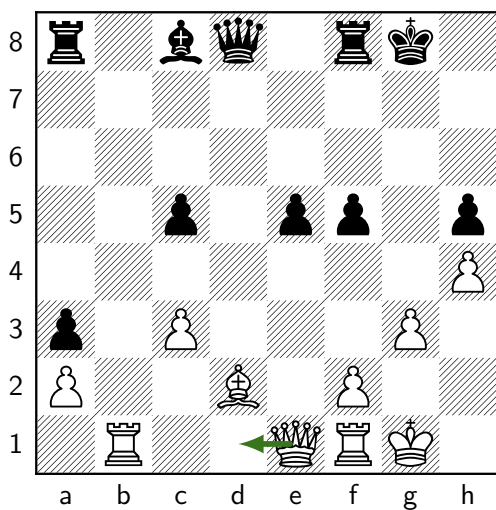


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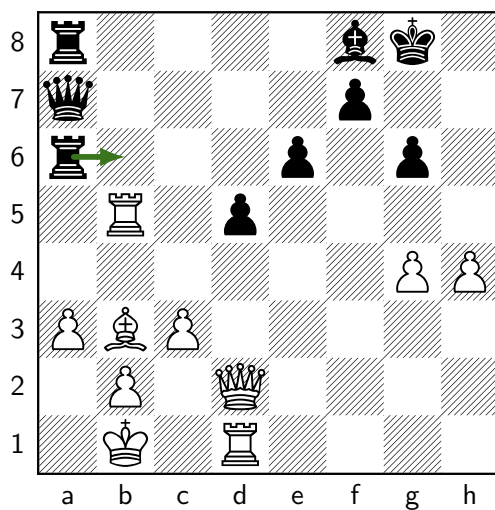


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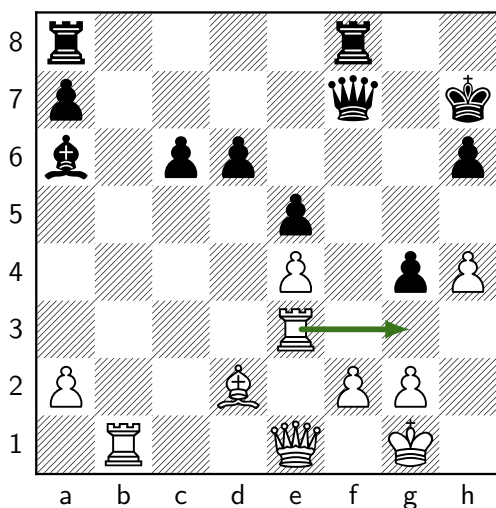
Figure 18: **Move 24, White: exf5.** (a) Our position; Melanie plays exf5 (e4f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#845 (ply 47) [structural distance $d=57$, QAP obj=60]: the master played d4; under μ our Pe_4 plays the role of that game’s Pd_4 ; cross-type roles: our Pe_4 as their Rf_8 , our Pg_3 as their Be_7 ; transferred move e4f5 (**endorses this move**). (c) Retrieved from ccr1_top_p0003#542 (ply 55) [structural distance $d=65$, QAP obj=58]: the master played exd5; under μ our Pe_4 plays the role of that game’s Pe_3 ; cross-type roles: our Qe_1 as their Rc_8 , our Pg_3 as their Qe_7 ; transferred move e4f5 (**endorses this move**). (d) Retrieved from ccr1_top_p0001#3385 (ply 46) [structural distance $d=68$, QAP obj=56]: the master played Bxa6; under μ our Pe_4 plays the role of that game’s Bd_3 (a **cross-type** correspondence); cross-type roles: our Ph_4 as their Rb_2 , our Rb_1 as their Ph_3 ; transferred move e4f5 (**endorses this move**). Synthesis: 13 retrieved analogs transferred to exf5 (summed QAP objective 495, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



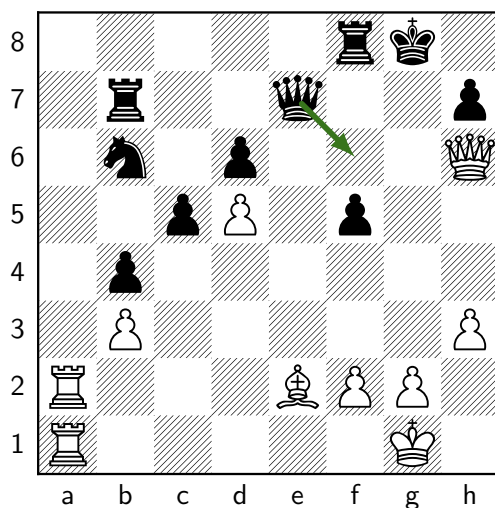
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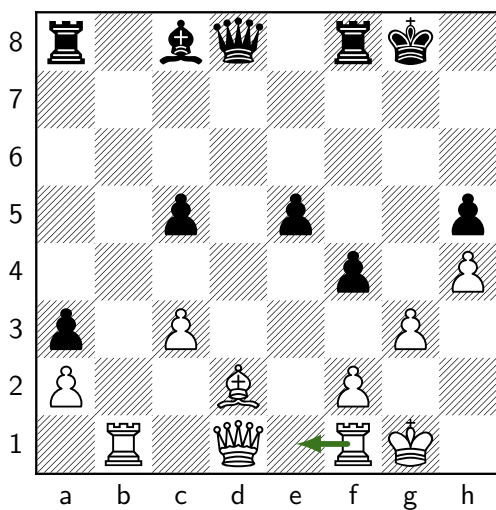


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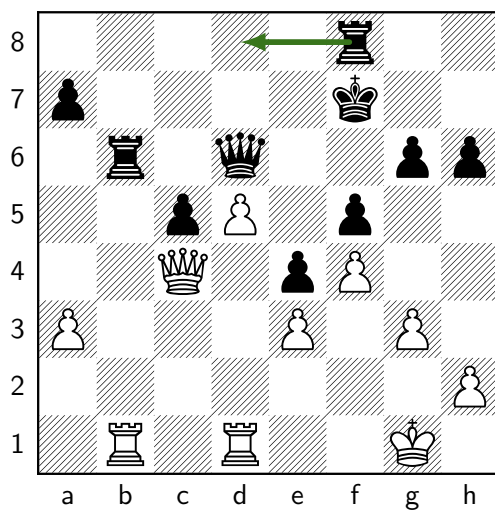


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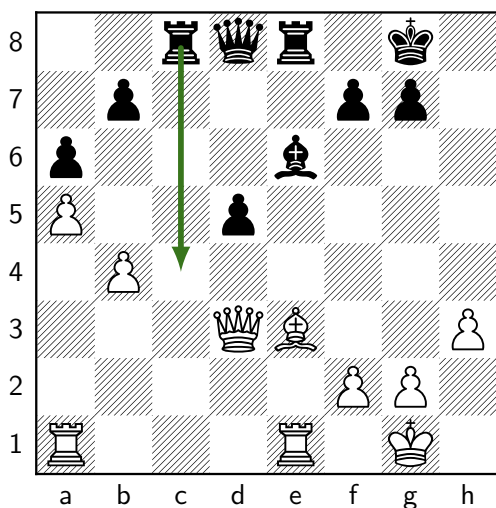
Figure 19: **Move 25, White: Qd1**. (a) Our position; Melanie plays Qd1 (e1d1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#2526` (ply 51) [structural distance $d=62$, QAP obj=62]: the master played Rb6; under μ our Qe1 plays the role of that game’s Ra3 (a **cross-type** correspondence); cross-type roles: our Qe1 as their Bf8, our Rb1 as their Qa7; transferred move e1d1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#568` (ply 52) [structural distance $d=45$, QAP obj=20]: the master played Rg3; under μ our Qe1 plays the role of that game’s Re3 (a **cross-type** correspondence); cross-type roles: our Pg3 as their Rb1, our Rf1 as their Qe1; transferred move e1d1 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#1428` (ply 51) [structural distance $d=68$, QAP obj=6]: the master played Qf6; under μ our Qe1 plays the role of that game’s Qe2; cross-type roles: our ra8 as their Nb6, our Rb1 as their Pd6; transferred move e1d1 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Qd1 (summed QAP objective 88, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



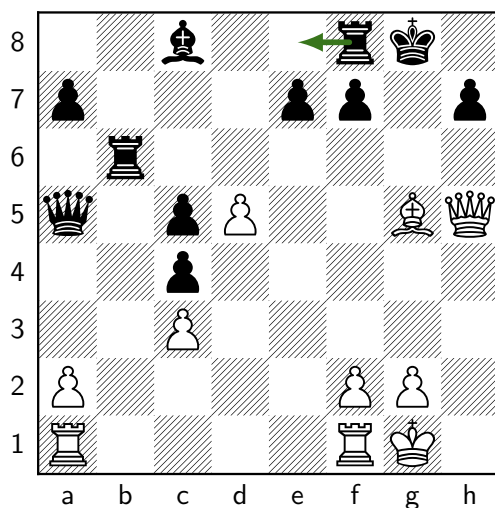
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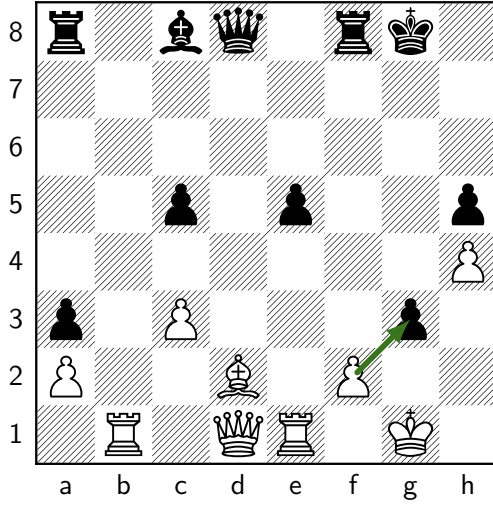


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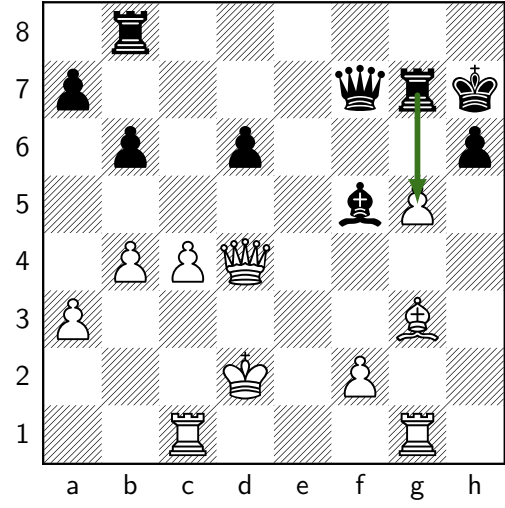


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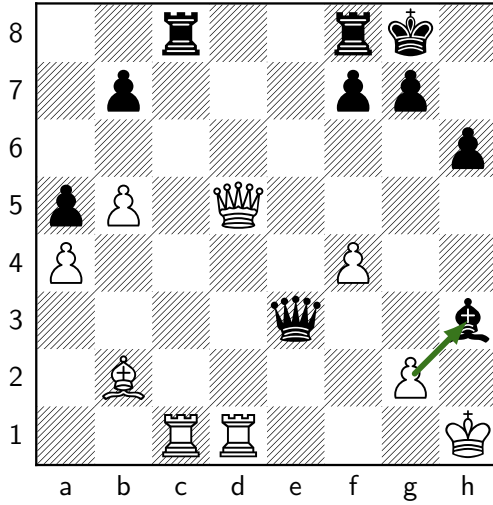
Figure 20: **Move 26, White: Re1**. (a) Our position; Melanie plays Re1 (f1e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#742` (ply 67) [structural distance $d=73$, QAP obj=81]: the master played Rd8; under μ our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *pf4* as their *Rb6*, our *rf8* as their *Pg6*; transferred move f1e1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#755` (ply 43) [structural distance $d=68$, QAP obj=56]: the master played Rc4; under μ our *Rf1* plays the role of that game’s *Rc1*; cross-type roles: our *bc8* as their *Pg7*, our *Pg3* as their *Be6*; transferred move f1e1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2482` (ply 35) [structural distance $d=68$, QAP obj=50]: the master played Re8; under μ our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *pc5* as their *Bc8*, our *rf8* as their *Pa7*; transferred move f1e1 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Re1 (summed QAP objective 336, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



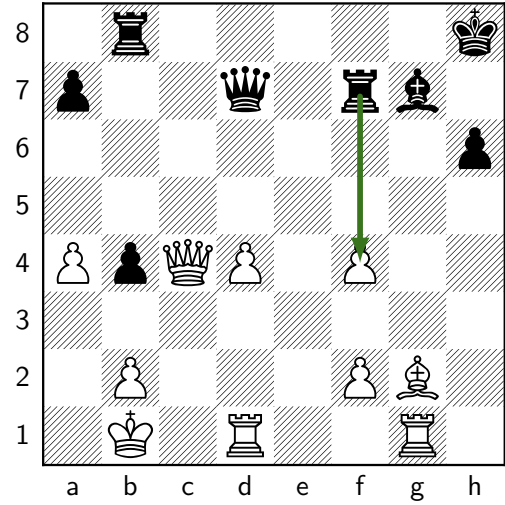
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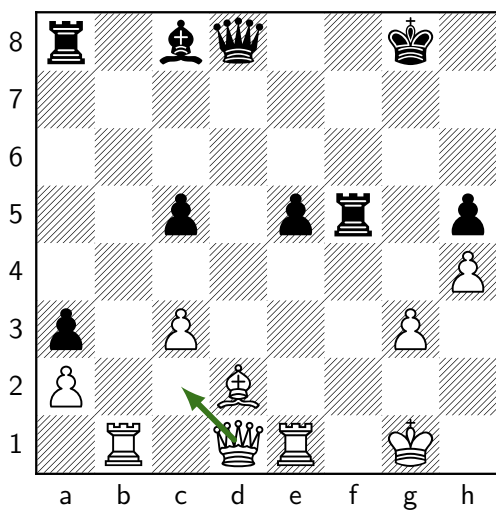


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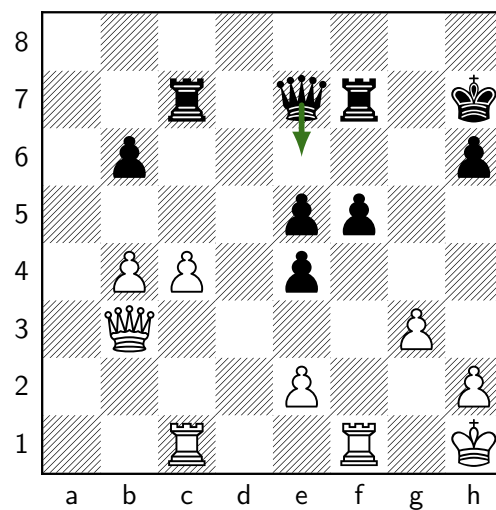


(d)

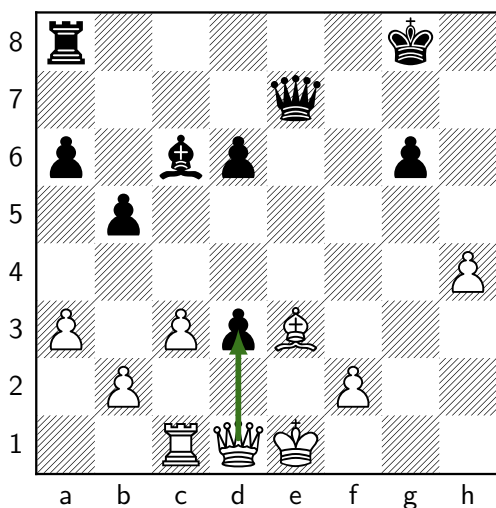
Figure 21: **Move 27, White: f2g3.** (a) Our position; Melanie plays f2g3 (f2g3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3377` (ply 57) [structural distance $d=75$, QAP obj=50]: the master played **Rxg5**; under μ our *Pf2* plays the role of that game’s *Rg2* (a **cross-type** correspondence); cross-type roles: our *Bd2* as their *Pd6*, our *Re1* as their *Bf5*; transferred move **f2g3** (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#416` (ply 52) [structural distance $d=83$, QAP obj=40]: the master played **gxh3**; under μ our *Pf2* plays the role of that game’s *Pg2*; cross-type roles: our *Pf2* as their *Rc1*, our *Pc3* as their *Bb2*; transferred move **f2g3** (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1429` (ply 53) [structural distance $d=81$, QAP obj=34]: the master played **Rxf4**; under μ our *Pf2* plays the role of that game’s *Rf2* (a **cross-type** correspondence); cross-type roles: our *ph5* as their *Rb8*, our *bc8* as their *Pa7*; transferred move **f2g3** (**endorses this move**). Synthesis: 3 retrieved analogs transferred to **f2g3** (summed QAP objective 124, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



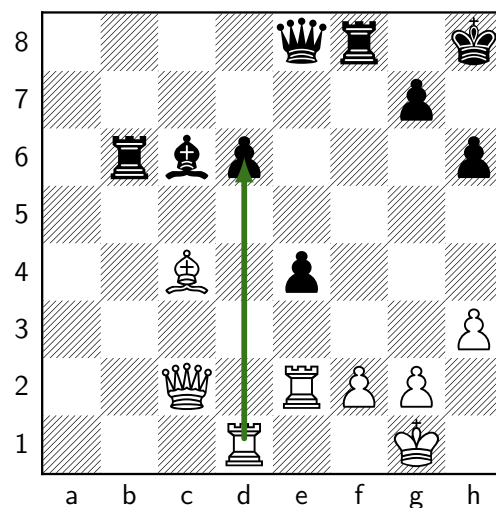
(a)



(b)

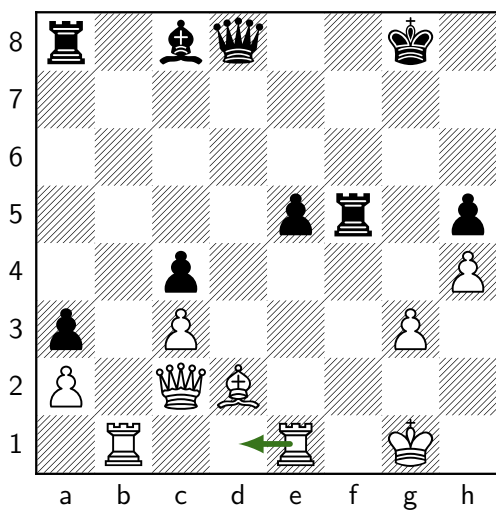


(c)

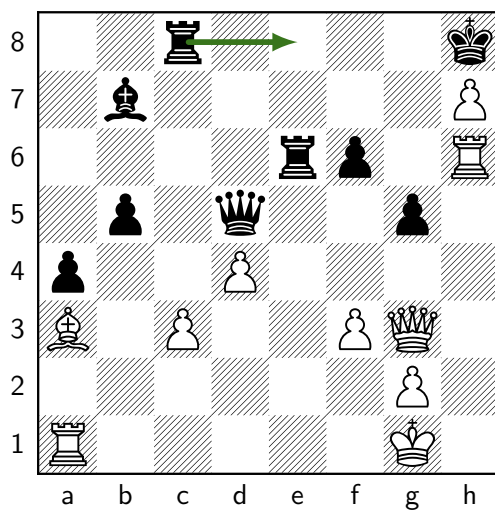


(d)

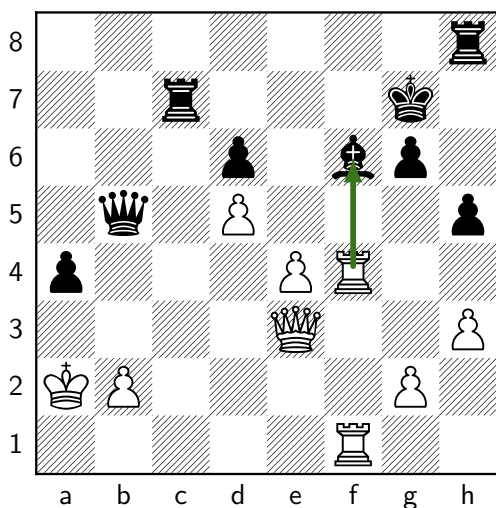
Figure 22: **Move 28, White: Qc2.** (a) Our position; Melanie plays Qc2 (d1c2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#416` (ply 65) [structural distance $d=83$, QAP obj=57]: the master played Qe6; under μ our Qd1 plays the role of that game’s Qe2; cross-type roles: our qd8 as their Rc7, our Rb1 as their Ph6; transferred move d1c2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2984` (ply 50) [structural distance $d=86$, QAP obj=49]: the master played Qxd3; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our bc8 as their Pb2, our ra8 as their Pc3; transferred move d1c2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#267` (ply 68) [structural distance $d=79$, QAP obj=69]: the master played Rxd6; under μ our Re1 plays the role of that game’s Rd1; cross-type roles: our qd8 as their Re2, our Bd2 as their pe4; transferred move e1e5. Synthesis: 2 retrieved analogs transferred to Qc2 (summed QAP objective 106, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



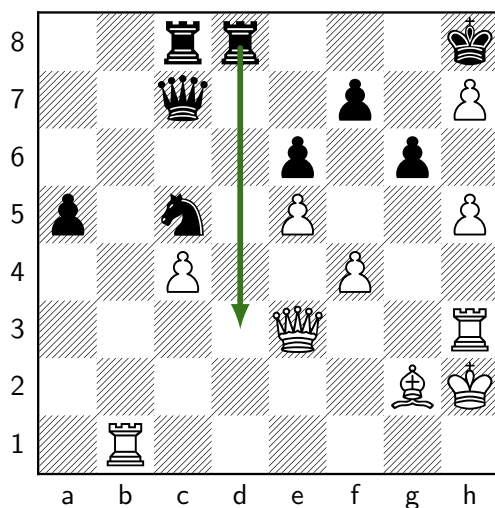
(a)



(b)

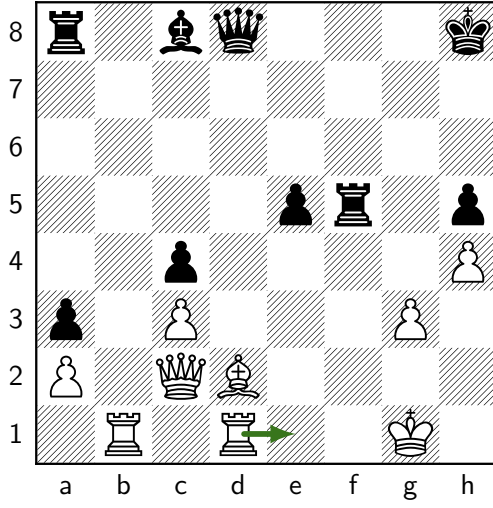


(c)

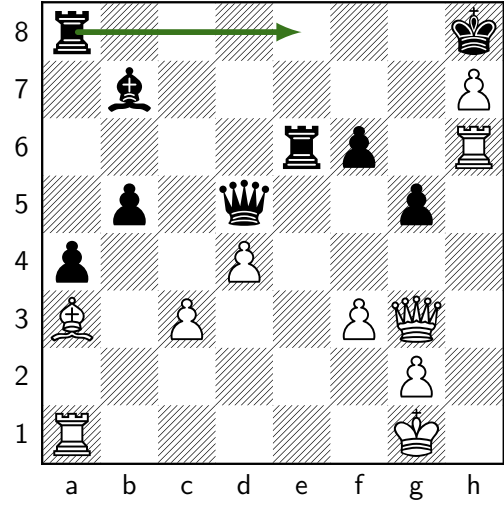


(d)

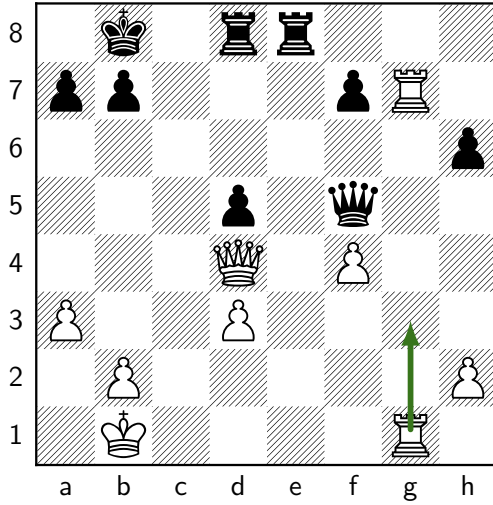
Figure 23: **Move 29, White: Red1.** (a) Our position; Melanie plays Red1 (e1d1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#479` (ply 67) [structural distance $d=79$, QAP obj=20]: the master played Rce8; under μ our *Re1* plays the role of that game’s *Rc1*; cross-type roles: our *pe5* as their *Bb7*, our *Pc3* as their *Re6*; transferred move **e1d1 (endorses this move)**. (c) Retrieved from `ccr1_top_p0000#3245` (ply 70) [structural distance $d=81$, QAP obj=29]: the master played Rxf6; under μ our *Qc2* plays the role of that game’s *Rf4* (a **cross-type** correspondence); cross-type roles: our *Bd2* as their *Rf1*, our *ra8* as their *Qe3*; transferred move **c2f5**. (d) Retrieved from `ccr1_top_p0000#2866` (ply 65) [structural distance $d=79$, QAP obj=28]: the master played Rd3; under μ our *Re1* plays the role of that game’s *Rd1*; cross-type roles: our *pa3* as their *Rc8*, our *Rb1* as their *Pa5*; transferred move **e1e4**. Synthesis: 3 retrieved analogs transferred to Red1 (summed QAP objective 61, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



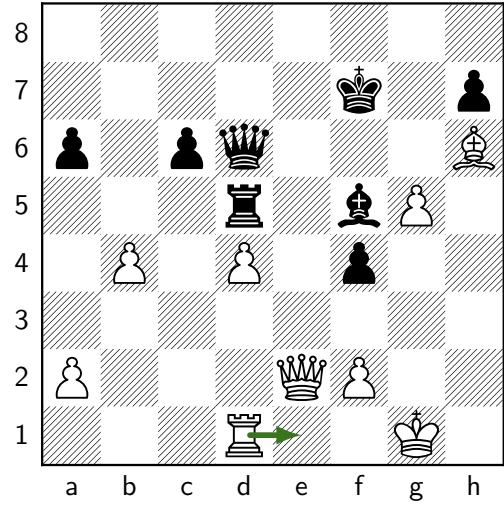
(a)



(b)

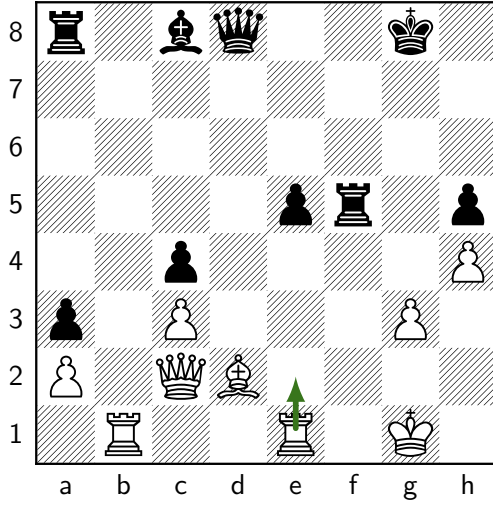


(c)

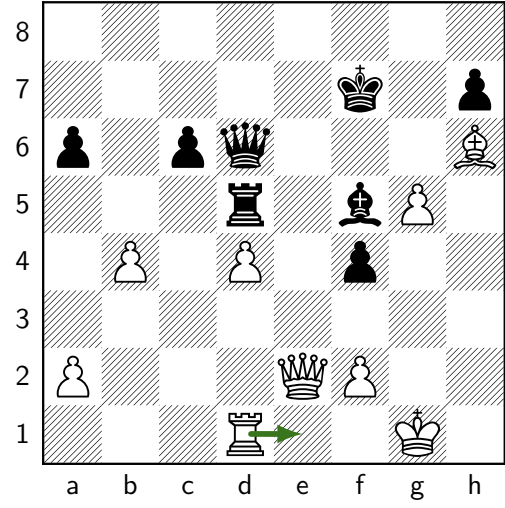


(d)

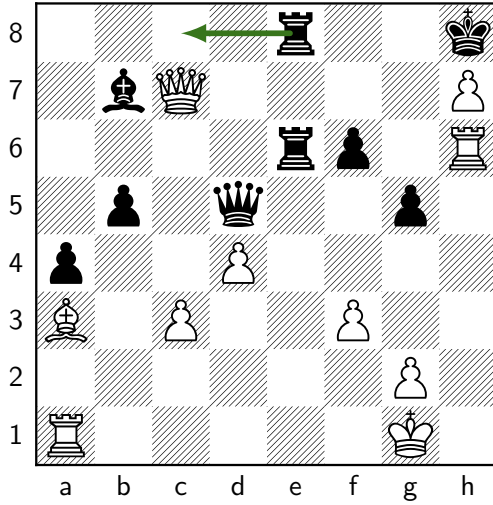
Figure 24: **Move 30, White: Re1**. (a) Our position; Melanie plays Re1 (d1e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#479` (ply 63) [structural distance $d=89$, QAP obj=40]: the master played Rae8; under μ our R $d1$ plays the role of that game’s R $a1$; cross-type roles: our P $c3$ as their B $b7$, our $bc8$ as their $ph7$; transferred move d1e1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2125` (ply 54) [structural distance $d=86$, QAP obj=35]: the master played R1g3; under μ our R $d1$ plays the role of that game’s R $g1$; cross-type roles: our R $b1$ as their P $h2$, our B $d2$ as their P $d3$; transferred move d1e1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2606` (ply 68) [structural distance $d=89$, QAP obj=35]: the master played Re1; under μ our R $d1$ plays the role of that game’s R $d1$; cross-type roles: our R $b1$ as their P $a2$, our $pc4$ as their Q $e2$; transferred move d1e1 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Re1 (summed QAP objective 189, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



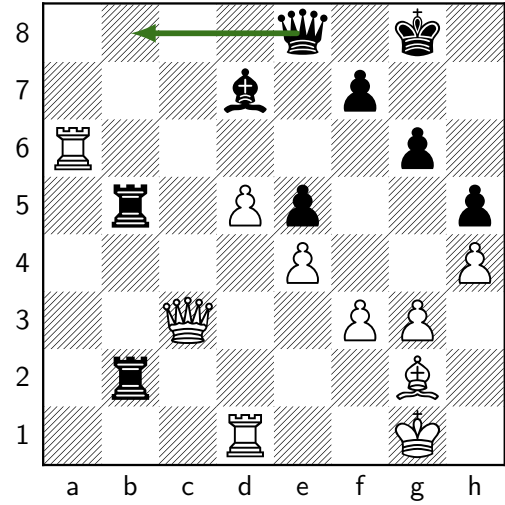
(a)



(b)

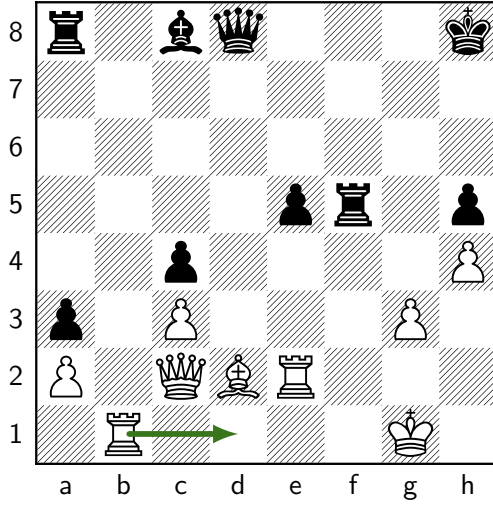


(c)

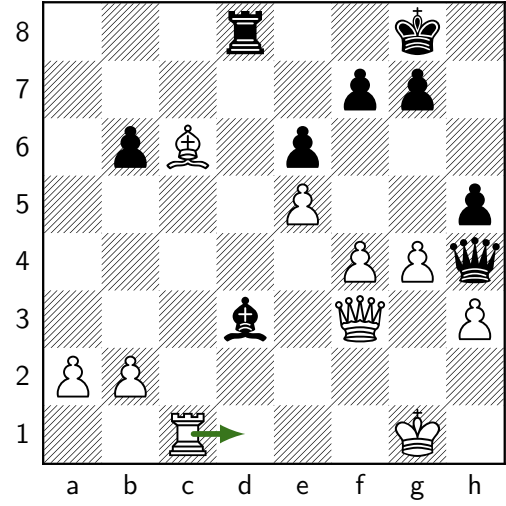


(d)

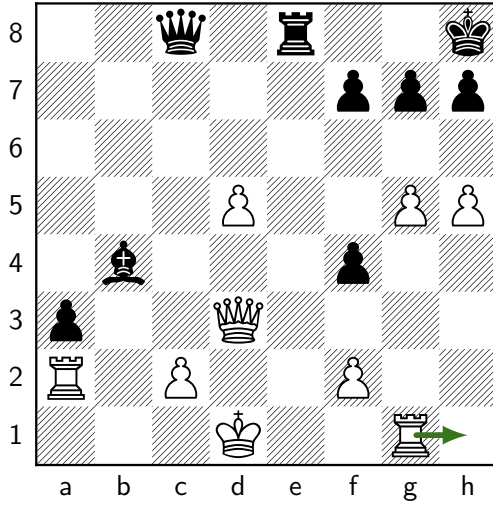
Figure 25: **Move 31, White: Re2.** (a) Our position; Melanie plays Re2 (e1e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccl1_top_p0003#2606` (ply 68) [structural distance $d=75$, QAP obj=21]: the master played Re1; under μ our Re1 plays the role of that game’s Rd1; cross-type roles: our Rb1 as their Pa2, our pc4 as their Qe2; transferred move e1e2 (**endorses this move**). (c) Retrieved from `ccl1_top_p0001#479` (ply 65) [structural distance $d=66$, QAP obj=14]: the master played Rc8; under μ our Re1 plays the role of that game’s Re1; cross-type roles: our rf5 as their Bb7, our Rb1 as their ph7; transferred move e1e2 (**endorses this move**). (d) Retrieved from `ccl1_top_p0001#1296` (ply 53) [structural distance $d=70$, QAP obj=10]: the master played Qb8; under μ our Re1 plays the role of that game’s Qe1 (a **cross-type** correspondence); cross-type roles: our pe5 as their ra6, our Rb1 as their Pg6; transferred move e1e2 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Re2 (summed QAP objective 59, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



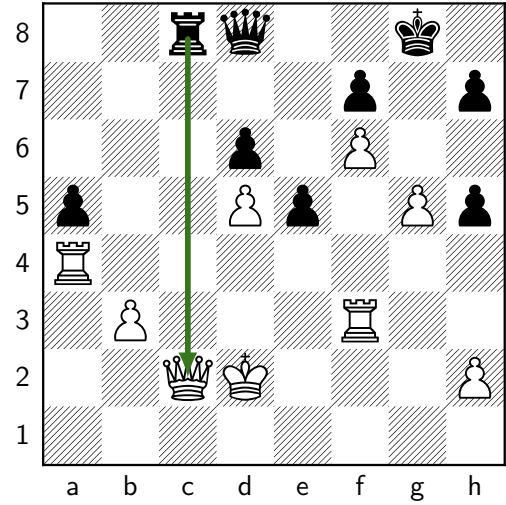
(a)



(b)

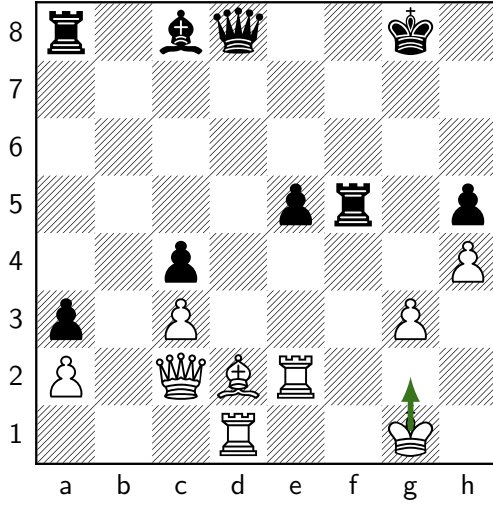


(c)

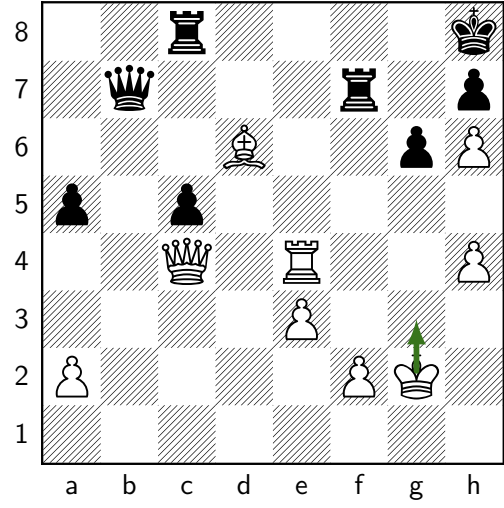


(d)

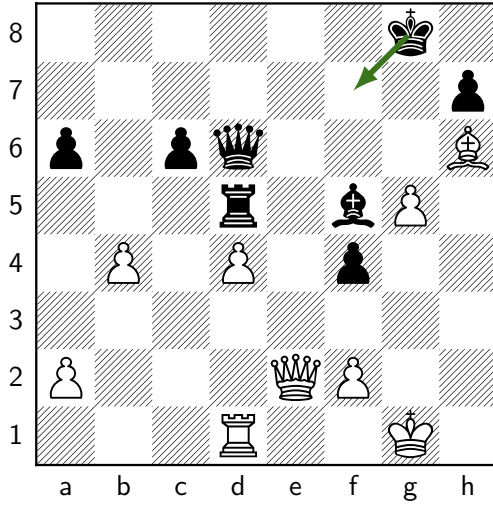
Figure 26: **Move 32, White: Rd1.** (a) Our position; Melanie plays Rd1 (b1d1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#922` (ply 56) [structural distance $d=65$, QAP obj=36]: the master played Rd1; under μ our Rb1 plays the role of that game’s Rc1; cross-type roles: our Re2 as their Pb2, our rf5 as their bd3; transferred move b1d1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3019` (ply 64) [structural distance $d=73$, QAP obj=15]: the master played Rh1; under μ our Rb1 plays the role of that game’s Rg1; cross-type roles: our Qc2 as their Ra2, our rf5 as their pa3; transferred move b1d1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#139` (ply 65) [structural distance $d=77$, QAP obj=27]: the master played Rxc2+; under μ our Qc2 plays the role of that game’s Rc1 (a **cross-type** correspondence); cross-type roles: our Bd2 as their pf6, our rf5 as their Pa5; transferred move c2a4. Synthesis: 2 retrieved analogs transferred to Rd1 (summed QAP objective 51, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



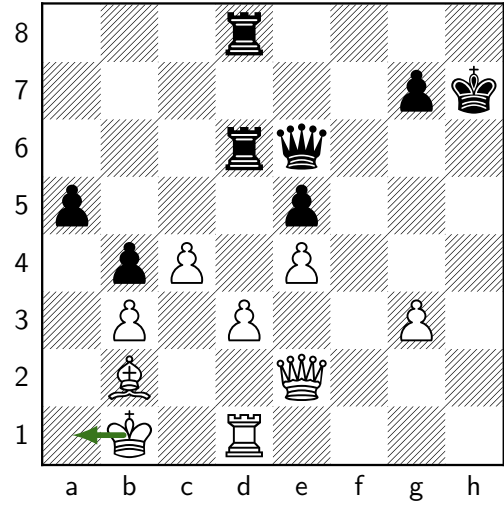
(a)



(b)

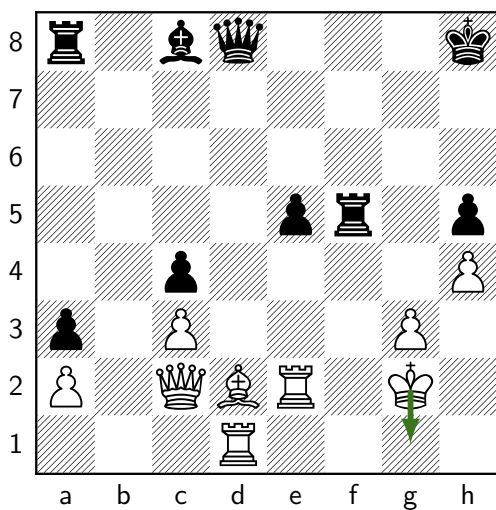


(c)

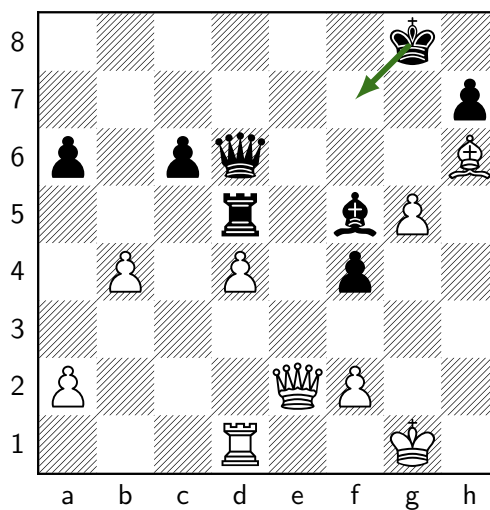


(d)

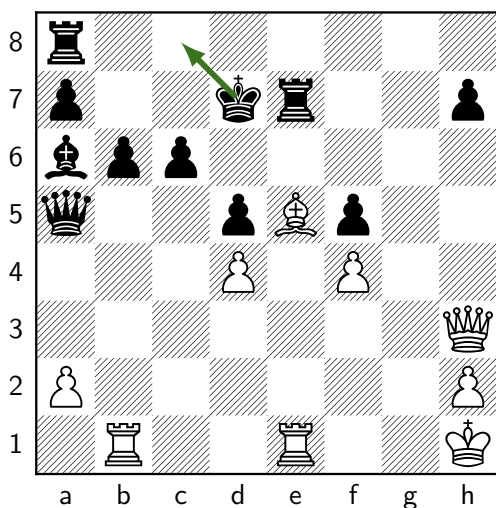
Figure 27: **Move 33, White: Kg2.** (a) Our position; Melanie plays Kg2 (**g1g2**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3914` (ply 68) [structural distance $d=81$, QAP obj=52]: the master played Kg3; under μ our Kg1 plays the role of that game’s Kg2; cross-type roles: our Rd1 as their Pa2, our Re2 as their Pe3; transferred move **g1g2 (endorses this move)**. (c) Retrieved from `ccrl_top_p0003#2606` (ply 67) [structural distance $d=75$, QAP obj=34]: the master played Kf7; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our rf5 as their bh6, our qd8 as their Rd5; transferred move **g1g2 (endorses this move)**. (d) Retrieved from `ccrl_top_p0001#260` (ply 76) [structural distance $d=74$, QAP obj=28]: the master played Ka1; under μ our Kg1 plays the role of that game’s Kb1; cross-type roles: our pe5 as their Bb2, our Re2 as their Qe2; transferred move **g1g2 (endorses this move)**. Synthesis: 4 retrieved analogs transferred to Kg2 (summed QAP objective 117, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



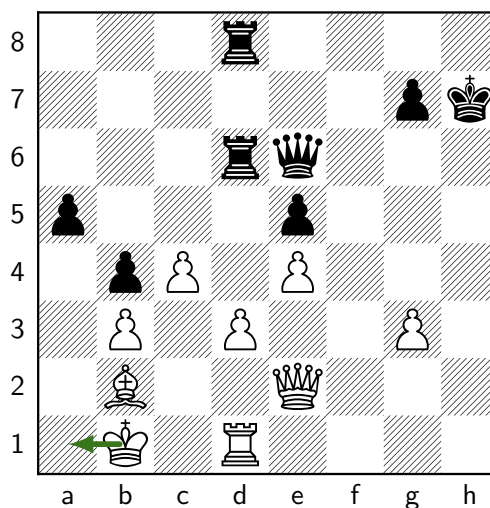
(a)



(b)

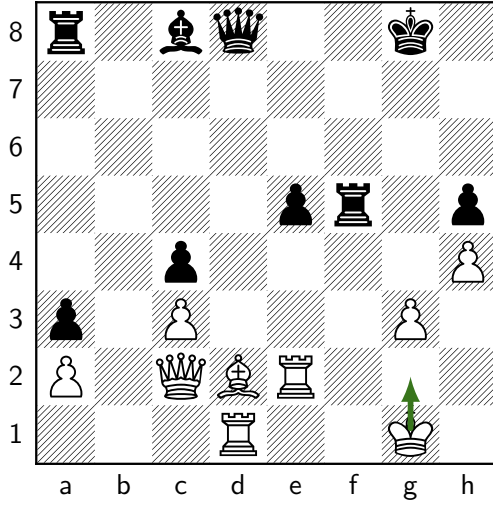


(c)

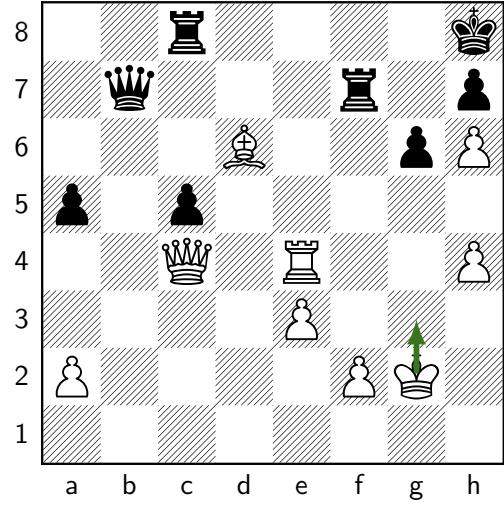


(d)

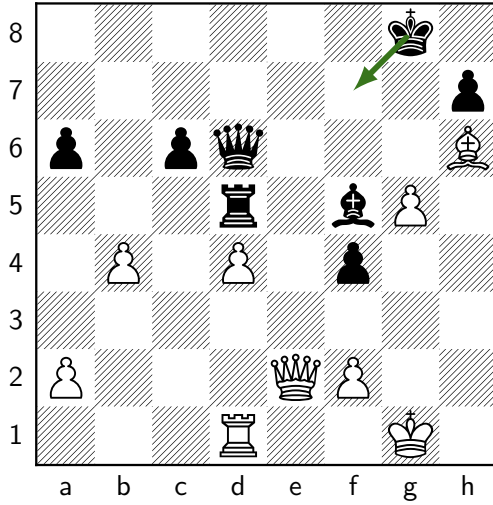
Figure 28: **Move 34, White: Kg1**. (a) Our position; Melanie plays Kg1 (**g2g1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2606` (ply 67) [structural distance $d=73$, QAP obj=33]: the master played Kf7; under μ our Kg2 plays the role of that game's Kg1; cross-type roles: our rf5 as their bh6, our qd8 as their Rd5; transferred move g2g1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2083` (ply 65) [structural distance $d=75$, QAP obj=32]: the master played Kc8; under μ our Kg2 plays the role of that game's Kd2; cross-type roles: our Qc2 as their Re7, our Pg3 as their Ba6; transferred move g2g1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#260` (ply 76) [structural distance $d=70$, QAP obj=28]: the master played Ka1; under μ our Kg2 plays the role of that game's Kb1; cross-type roles: our pe5 as their Bb2, our Rd1 as their Qe2; transferred move g2g1 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kg1 (summed QAP objective 99, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



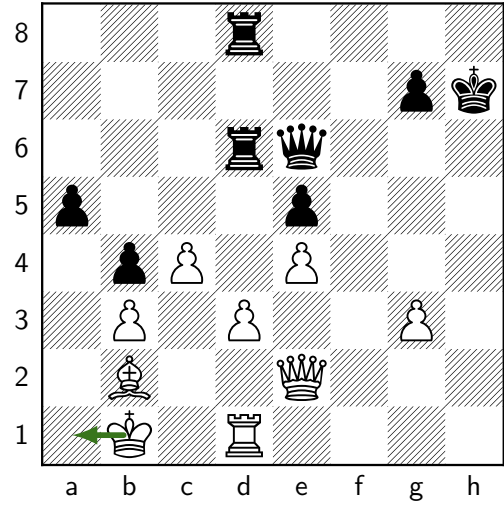
(a)



(b)



(c)



(d)

Figure 29: **Move 35, White: Kg2.** (a) Our position; Melanie plays Kg2 (**g1g2**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3914` (ply 68) [structural distance $d=81$, QAP obj=52]: the master played Kg3; under μ our Kg1 plays the role of that game’s Kg2; cross-type roles: our Rd1 as their Pa2, our Re2 as their Pe3; transferred move **g1g2 (endorses this move)**. (c) Retrieved from `ccrl_top_p0003#2606` (ply 67) [structural distance $d=75$, QAP obj=34]: the master played Kf7; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our rf5 as their bh6, our qd8 as their Rd5; transferred move **g1g2 (endorses this move)**. (d) Retrieved from `ccrl_top_p0001#260` (ply 76) [structural distance $d=74$, QAP obj=28]: the master played Ka1; under μ our Kg1 plays the role of that game’s Kb1; cross-type roles: our pe5 as their Bb2, our Re2 as their Qe2; transferred move **g1g2 (endorses this move)**. Synthesis: 5 retrieved analogs transferred to Kg2 (summed QAP objective 143, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.